

DRC Design Verification Test

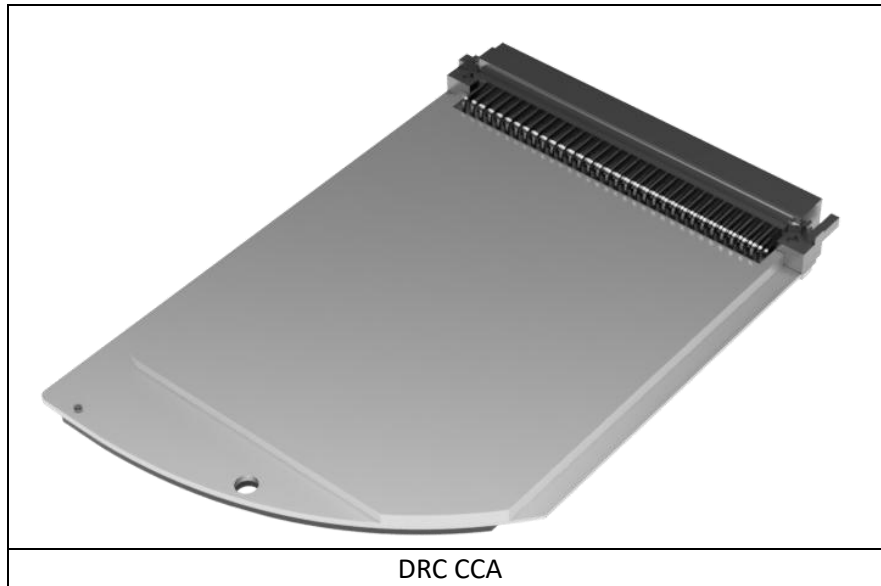
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INFORMATION FOR PROGRAM USE ONLY			

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Introduction

This DVT serves to provide the hardware verification procedure to test all circuitry contained within the DRC.



Hardware Support

The following part numbers are support by the highest level version of the DVT below.

Name	Part Number	DVT Version
Digital Replacement Card (DRC)	OC-202-1-03	2.1

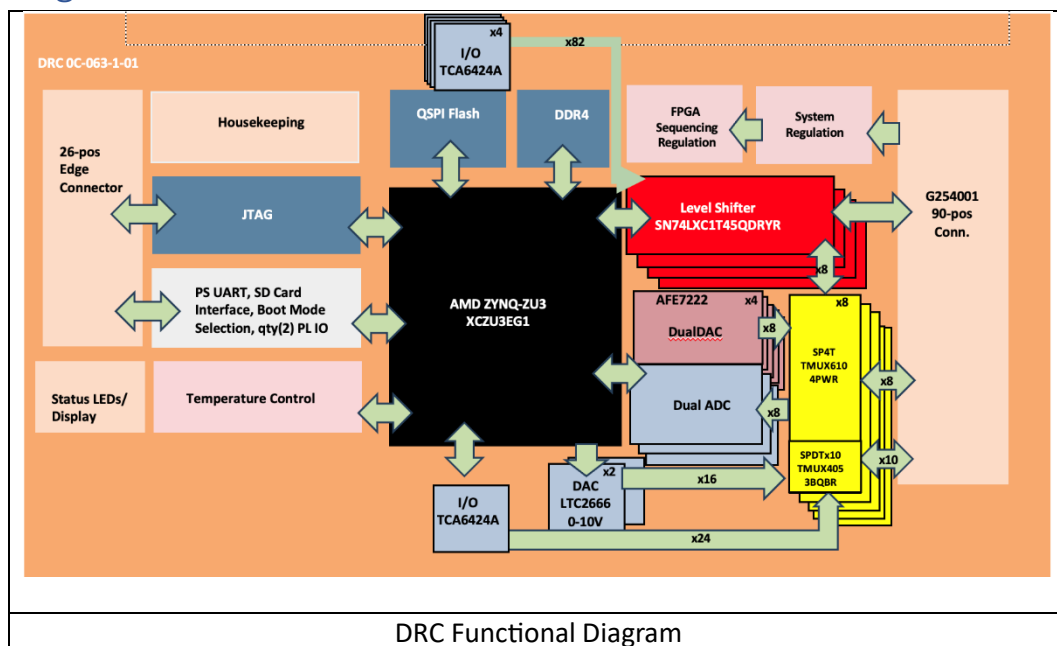
Changelog

Version	Date	Notes
Initial Draft Release 1.0	20 November 2024	Initial release
DRC Rev 2.1 Release 2.1	28 April 2025	Updated tests to match updated hardware.

Specifications

- Xilinx Zynq Ultra-Scale+ MPSoC ZU3, XCZU3EG1
- Quad, ARM® Cortex™-A53 Processors
- Dual, Arm Cortex R5 Real-Time Processors
- ARM Mail 400, GPU
- 8-Channel High Speed 65 MSPS ADCs
- 8-Channel High Speed 130 MSPS DACs
- 16-Channel Low Speed DACs , 0 to 10V
- 2 GB DDR4
- 128 MB quad SPI flash
- JTAG edge connector
- 90-pin G254001 Backplane Connector
- -40C to 85C Operation
- (+15V, -15V, +5V) OR (+5V only) Input
- Power dissipation between 5W and 8W (12W w/ 2 HSDAC channels active)

DRC Block Diagram



Equipment Required for Testing

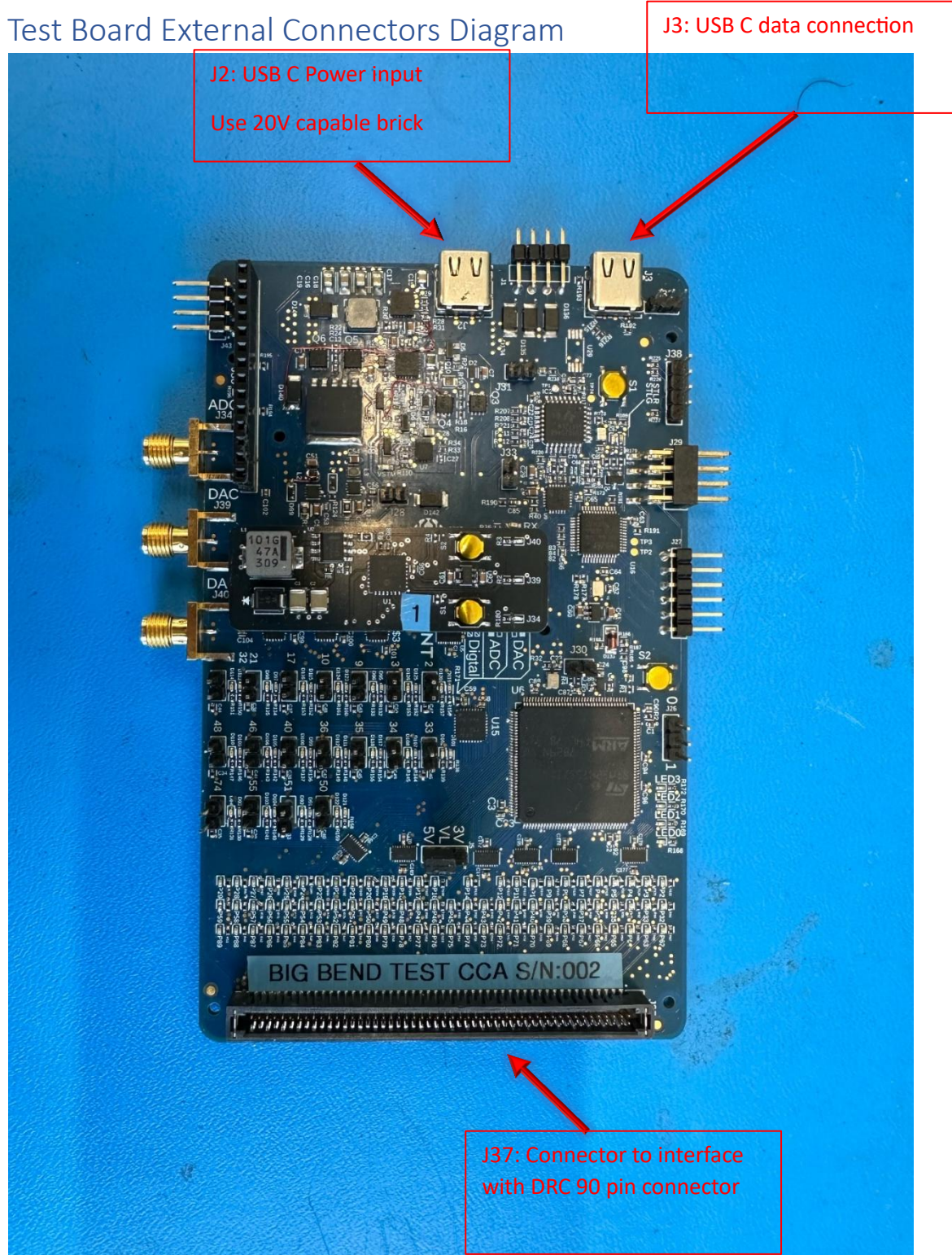
Test assembly:

1) DRC Rev. 2.1	OC-202-1-03
2) Test Board	OC-203-1-01
3) Prototype Chassis	OC-204-1-01
4) Test Board to DVT Interposer Board Cable	OC-212-1-01
5) DVT Interposer Board	OC-205-1-01
6) Power Brick	OC-206-1-01
7) USB C to USB A Cable (x2)	OC-207-1-01
8) USB C to USB C Cable	OC-208-1-01
9) DRC edge connector Program/Debug Board	OC-210-1-01
10) DRC JTAG Programmer (Modified HS3 Cable)	OC-211-1-01

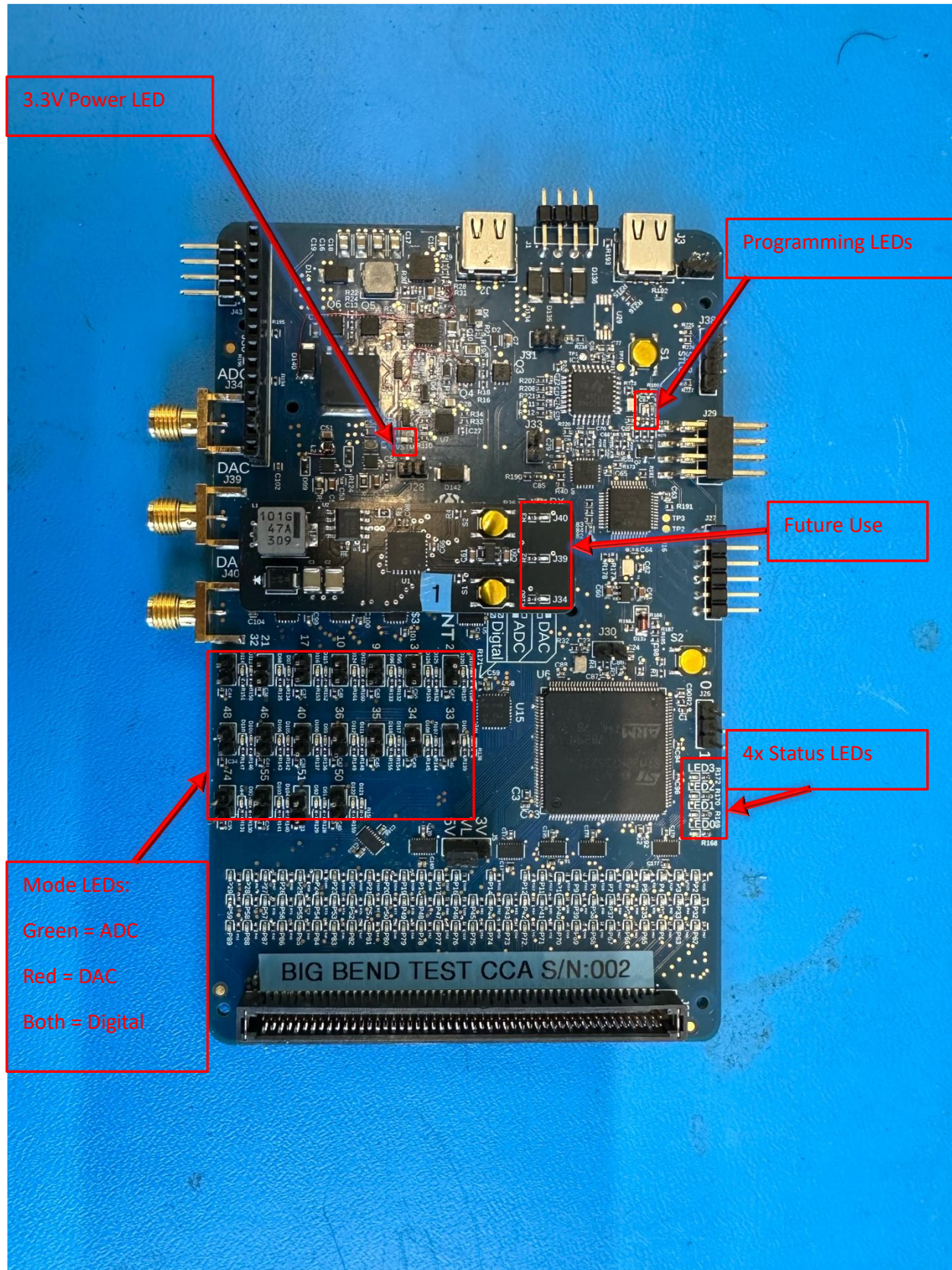
Equipment:

- 1) Signal generator
- 2) Oscilloscope
- 3) Computer
- 4) TeraTerm(or similar)
- 5) Vivado 2023.1
- 6) Vitis 2023.1

Test Board External Connectors Diagram



Test Board LED Diagram



J43: Internal Power Outputs
GND
-15V
15V
5V

J31: USB HUB RST

J1: External Power Inputs
5V | 15V | -15V | GND

J36: Reserved

J32: Reserved

J38: Reserved

J29: Programming Jumpers

J30: UART
TX RX

J27: ST Link SWD Connections
Do Not Use

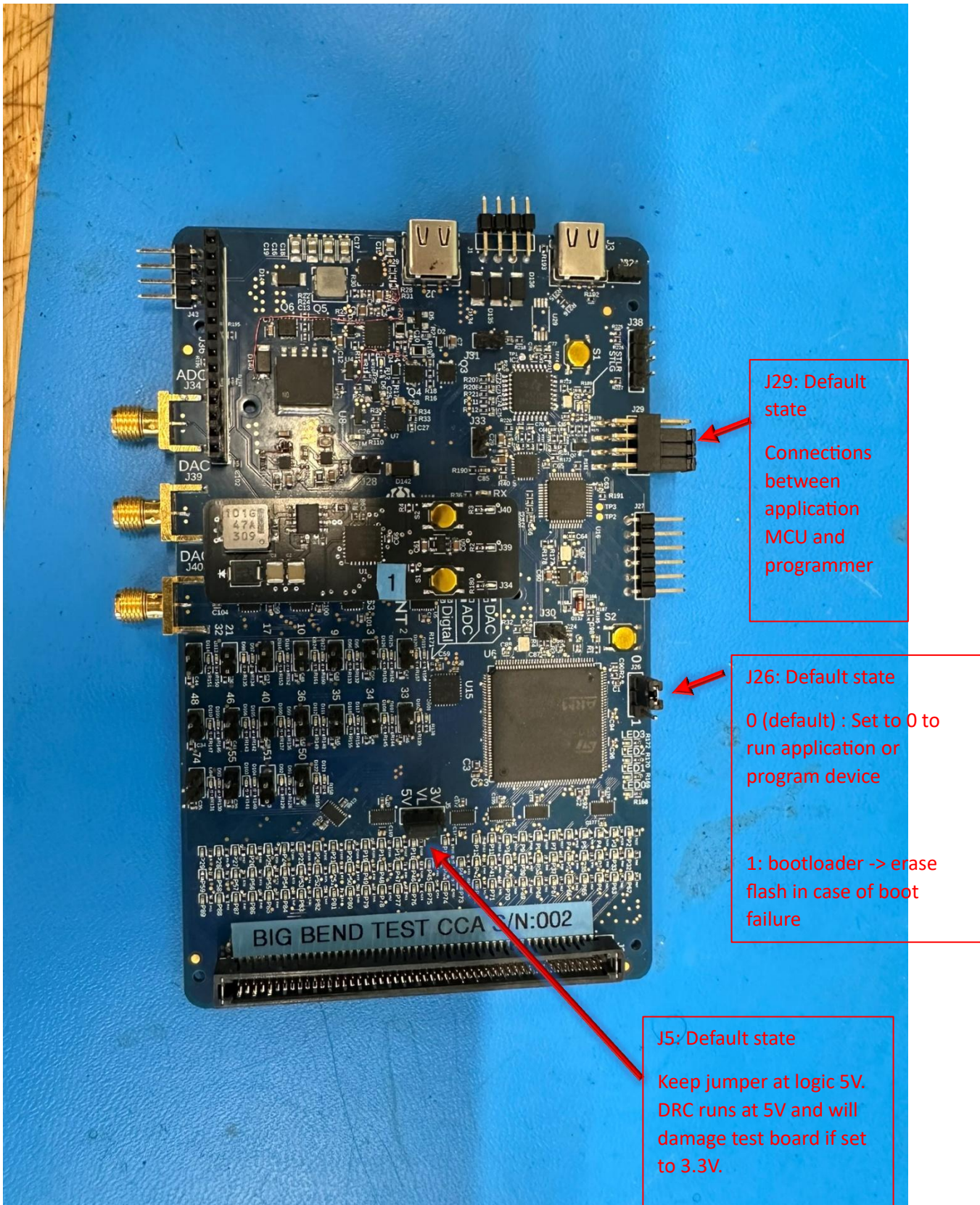
J26: Boot Mode Jumper

J5: Logic Voltage:
NOTE: KEEP AT 5V

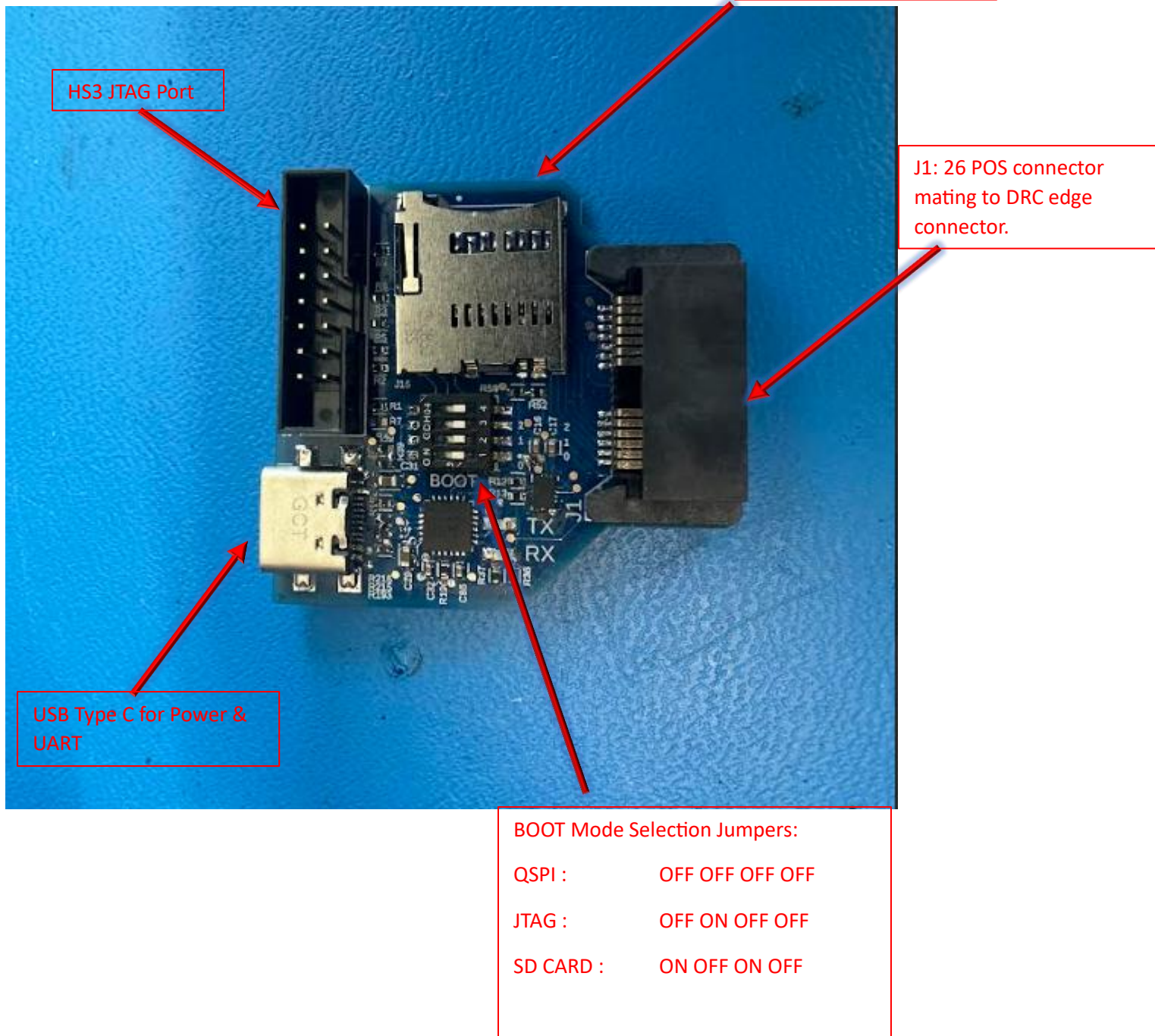
= not used

BIG BEND TEST CCA S/N:002

Test Board Default Jumper Placement

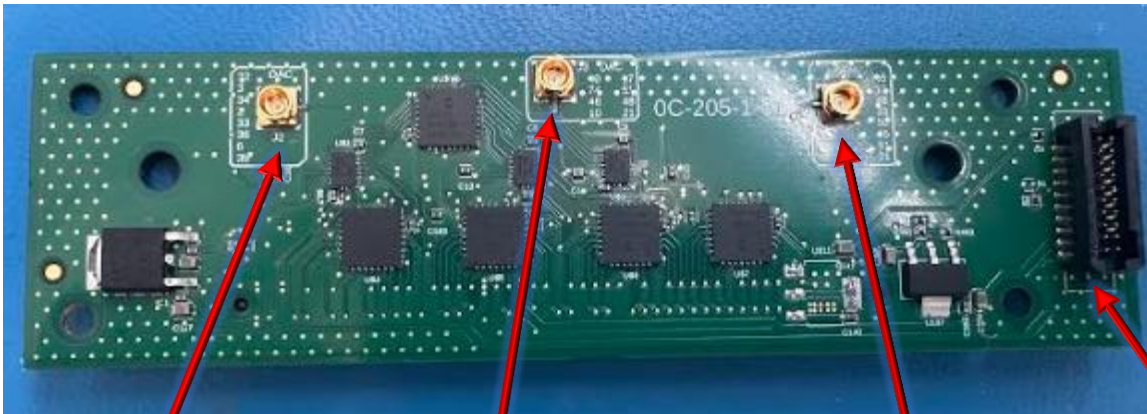
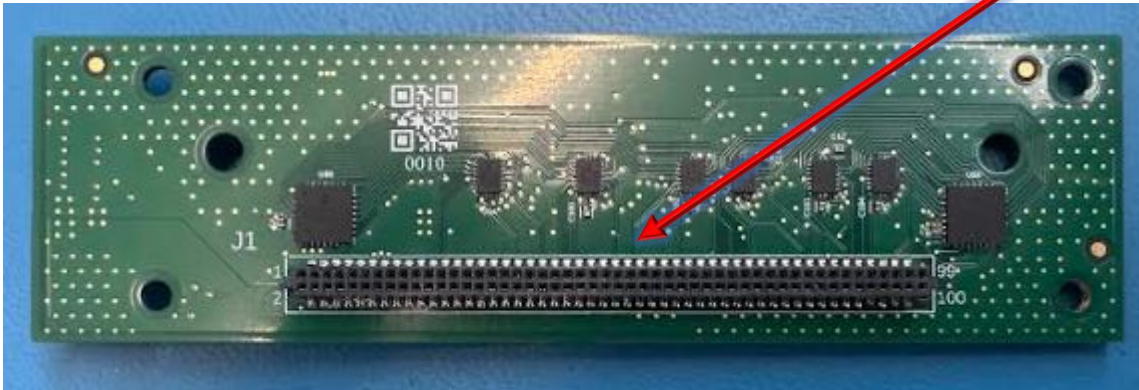


DRC Edge Connector Program/Debug Board



DVT Interposer Board

J1 : Connector to 90 pin interposer board.



J2 : DAC Output
SMPPM

Connects to Pins:

32
3
34
2
33
36
6
35

J3 : DAC Output
SMPPM

Connects to Pins:

40
74
46
10
17
51
48
21

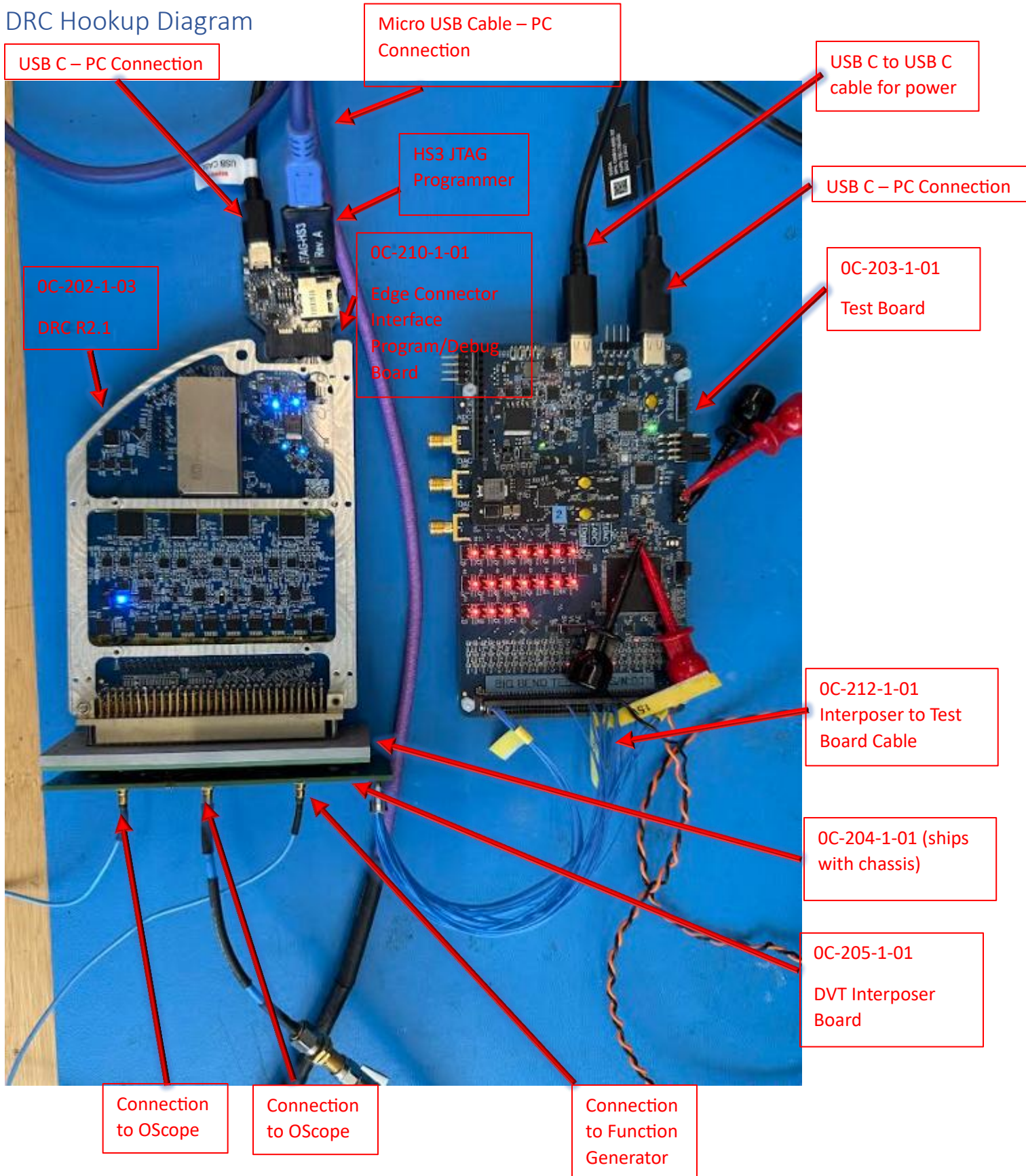
J4 : ADC Input
SMPPM

Connects to Pins:

46
17
48
33
60
55
51
21

20 POS Connector to Test
Board.

DRC Hookup Diagram



DRC Hookup Procedure

- 1) Connect the DVT Interposer Board to the 90-Pin Connector Interposer Board (this will be mounted to the chassis).
- 2) Connect the Ribbon Cable between the Test Board and the DVT Interposer Board
- 3) Slide the DRC into the Chassis and connect to the 90-Pin Connector Interposer board.
- 4) Make sure the Test Board and its Daughter Board have good connection. If this was removed and reseated, make sure the pin headers/connectors are properly aligned, and that all pins are in their sockets.
- 5) Connect the DRC edge connector Program/Debug Board to the DRC by mounting onto the DRC's edge connector opposite of the 90-pin connector.
- 6) Connect the Modified HS3 Programmer to the DRC edge connector Program/Debug Board.
 - a. Note: Can use unmodified HS3 programmer if USB C connected.
- 7) Connect the HS3 Programmer to a PC via a Micro USB to USB A cable.
- 8) Connect the DRC edge connector Program/Debug Board to a PC with a USB C to USB A cable.
- 9) Connect the Test Board to a PC with a USB C to USB A cable on J3.
- 10) Set the boot mode jumper to QSPI boot. BOOT DIP SWITCH -> (OFF OFF OFF OFF)
- 11) Connect J2 on the DVT Interposer Board to CH1 on an Oscilloscope through an SMPM to BNC cable.
- 12) Connect J3 on the DVT Interposer Board to CH3 on an Oscilloscope through an SMPM to BNC cable.
- 13) Connect J4 on the DVT Interposer Board to a function generator through an SMPM to BNC cable.
- 14) Connect the PD Power Brick to a receptacle.
- 15) Apply power to the system by connecting a USB C to USB C cable from the PD Power Brick to the Power connector on the Test Board.
- 16) Verify that the system powers up.
- 17) Check the edge mounted LEDs on the DRC for activity. With the default program, LED3 should be ON and LED1, LED2, and LED4 should be OFF on startup.

DRC Power Test and Verification Procedure

1) 90 Pin Connector

- a. Between 90-pin connector GND (Pin 1, 31, or 61) and 5V (Pin 30, 60, or 90), verify 5V is present
- b. Between 90-pin connector GND (Pin 1, 31, or 61) and 15V (Pin 13), verify 15V is present
- c. Between 90-pin connector GND (Pin 1, 31, or 61) and -15V (Pin 15), verify -12V is present. The test fixture provides -12V on this pin.

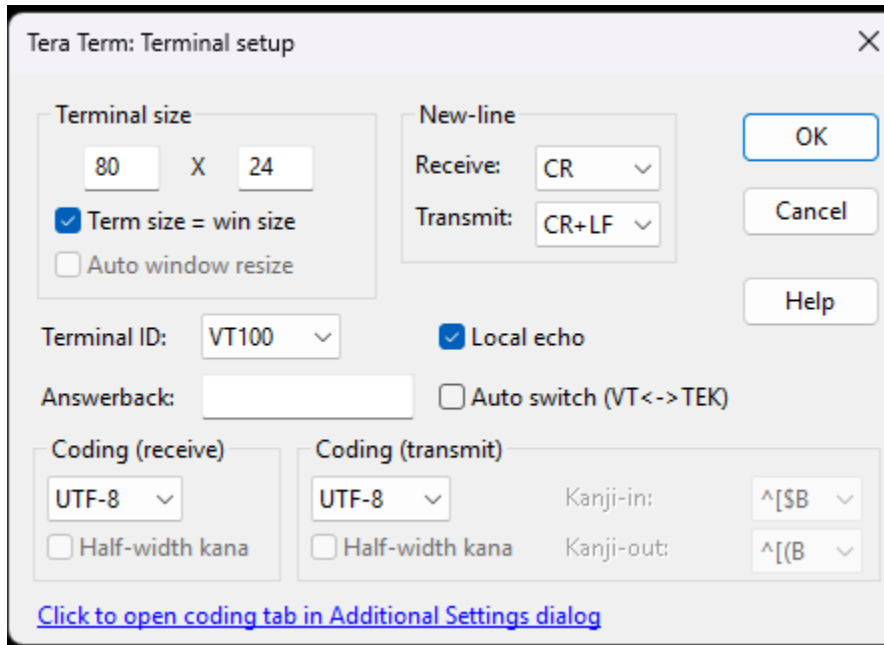
2) OSDZU3EG SiP Internal Regulation Test Points

1. Verify the following voltages at each test point in the table

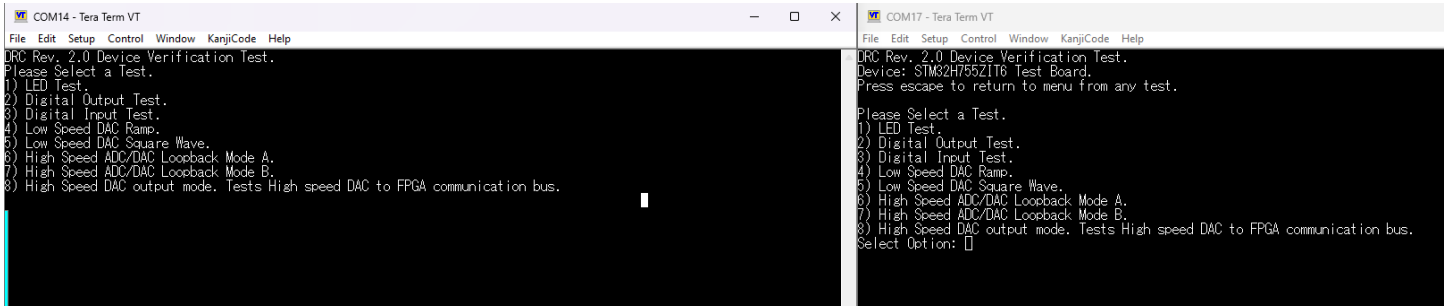
Test Point	Signal	Expected Voltage
TP22	V5P0	5.0V
TP23	PMIC2_C	3.3V
TP19	+3.3V_ON	3.3V
TP7	VCCINT	0.85V
TP8	VCC_PSINTLP	0.85V
TP9	VCC_PSINTFP	0.85V
TP15	PMIC1_B	1.8V
TP17	VCCAUX	1.8V
TP18	VCC_PSAUX	1.8V
TP31	AGND	0V

Connecting the DRC to a PC

- 1) Make sure the Edge Connector Interface Program/Debug Board and the Test Board are connected to a PC.
 - a. NOTE: You may get an unknown device error from the Test Board. This is OK as long as a USB to UART bridge is enumerated.
- 2) Open 2 sessions of TeraTerm and connect to both serial ports.
- 3) Set the baud rate of both sessions to 115200.
- 4) Navigate to Setup->Terminal and use these settings: Local Echo ON and Transmit is CR+LF.



- 5) Since the board has already booted, you will not see any terminal output. Remove power from the DRC and reapply power without closing the terminal session.
- 6) A list of tests will now be loaded into the terminals. The following section will describe each test. You should see this on the terminals. Left is the Test Board output and Right is the DRC output.



- 7) For each test, Both the Test Board and the DRC must be set to the same test mode. Operating different tests at the same time may cause unpredictable behavior. The DRC is line buffered, so ENTER must be pressed after sending commands, while the Test Board will act upon keyboard input in real time.

DRC Test Procedure Overview

For each test, the edge mounted LEDs on the DRC and the 4 status LEDs on the Test Board will indicate the current test. For example, in test mode 2, The DRC LEDs will illuminate 0010 and the Test Board LEDs will illuminate 0010. For both the Test Board and the DRC, the test mode must be selected using each board's

Test Mode 1: Cycles edge mounted LEDs on the DRC

- **Summary:** Outputs a pattern on the status LEDs
- **Verification:**
 - o Edge mounted LEDs and control circuitry

Test Mode 2: Set all pins to Digital paths and cycle IO pins.

- **Summary:** This test will first set all multipurpose pins' paths (SP4T or SPDT switches) to digital mode, whose signal path is FPGA (PL) -> Level Translator -> switch -> 90 pin connector. Afterward, the IO pins will turn on and off one at a time, ending with all IO pins being turned on for the duration of the runtime parameter.
- **Verification:**
 - o SPDT switch digital paths
 - o SP4T switch digital paths
 - o IO expander control
 - o Level shifters
 - o IO pins

Test Mode 3: Read IO Pins and report state changes.

- **Summary:** This test will first set all multipurpose pins' paths (SP4T or SPDT switches) to digital mode, whose signal path is 90 pin connector -> switch -> Level translator -> FPGA (PL). Afterward, the DRC will read each IO pin state on the 90 pin connector, and report all state changes on the pins. If the DRC reads at least one transition on each IO pin, a success message will be displayed. If not, the DRC will list out any pins that failed to transition.
- **Verification:**
 - o SPDT switch digital paths
 - o SP4T switch digital paths
 - o IO expander control
 - o Level shifters
 - o IO pins

Test Mode 4: Low speed DAC sawtooth wave outputs

- **Summary:** This test will first set all multipurpose pins' paths (SP4T or SPDT switches) to Low Speed DAC mode, whose signal path is Low Speed DAC channel -> switch -> 90 pin connector. Afterward, the DRC will send commands to all 16 Low Speed DAC channels to output a low frequency sawtooth wave.
- **Verification:**
 - o SPDT switch analog paths
 - o SP4T switch analog paths

- IO expander control
- Low Speed 8-channel DACs
- SPI control lines

Test Mode 5: Low speed DAC square wave outputs. Transition time test.

- **Summary:** This test will first set all multipurpose pins' paths (SP4T or SPDT switches) to Low Speed DAC mode, whose signal path is Low Speed DAC channel -> switch -> 90 pin connector. Afterward, the DRC will send commands to all 16 Low Speed DAC channels to cycle between 0V and 10V.
- **Verification:**
 - SPDT switch analog paths
 - SP4T switch analog paths
 - IO expander control
 - Low Speed 8-channel DACs
 - SPI control lines
 - < 2.5us transition time (10V/2.5us slew rate)

Test Mode 6: High speed ADC/DAC loopback mode A.

- **Summary:** Tests channel A on each ADC/DAC pair. Since some High Speed ADC and High Speed DAC channels share a multifunction pin, testing channel A and B for the converters will be done in separate tests.
- **Verification:**
 - SP4T switch analog paths
 - IO expander control
 - High Speed Converters (A Channels)

Test Mode 7: High speed ADC/DAC loopback mode B.

- **Summary:** Tests channel A on each ADC/DAC pair. Since some High Speed ADC and High Speed DAC channels share a multifunction pin, testing channel A and B for the converters will be done in separate tests.
- **Verification:**
 - SP4T switch analog paths
 - IO expander control
 - High Speed Converters (B Channels)

Test Mode 8: High speed DAC Output mode. DAC Output and Quadrature Generation.

- **Summary:** Tests all HSDAC outputs. During this test, the DACs will generate either a square wave (Mixing Disabled) or sine wave (Mixing enabled). This test tests 1 AFE DAC pair at a time.
- **Verification:**
 - SP4T switch analog paths
 - IO expander control
 - High Speed DACs
 - FPGA to AFE7222 communication

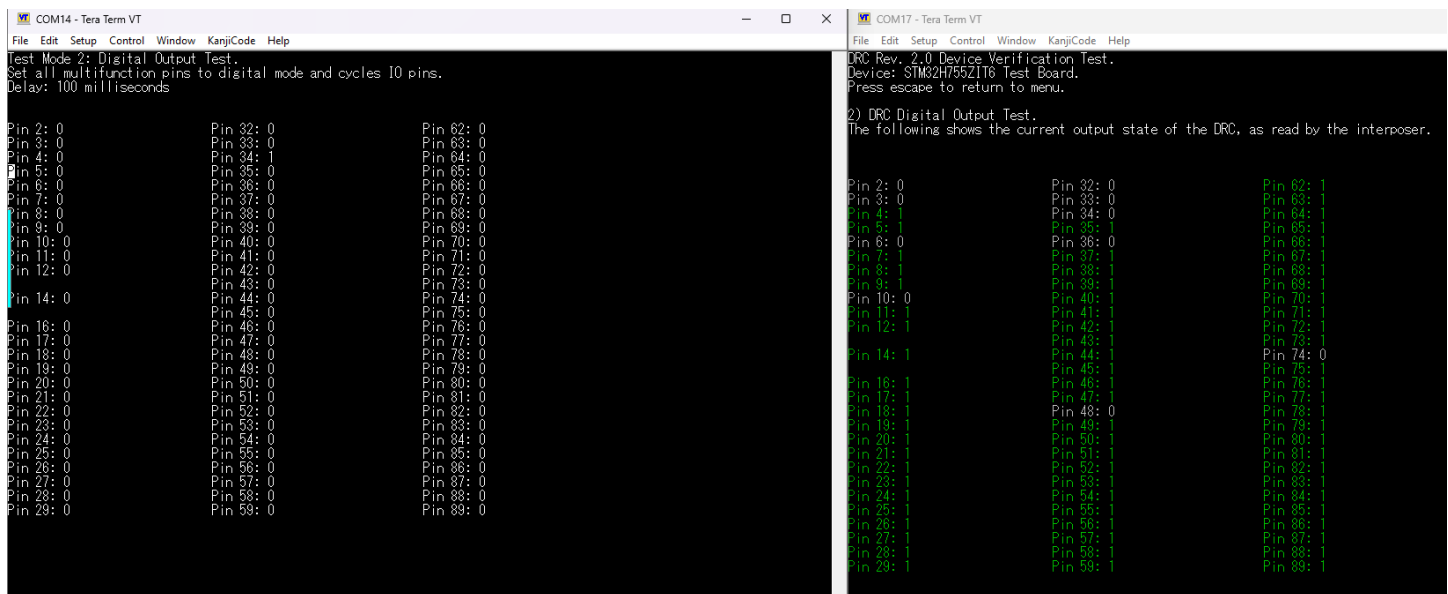
Test Mode 1: Set DRC Edge Mounted Status LEDs to 0001

- 1) When prompted to input a test mode, type “1” and press enter on the DRC console. No need to enter anything into the Test Board console.
- 2) The DRC status LEDs will cycle in the following order:
0001
0010
0100
1000
0100
0010
0001
- 3) Check that the status LEDs ends at 0001, indicating test mode 1. If so, the test is successful.

Test Mode 2: Set all pins to Digital paths and cycle IO pins.

This test will first set all multipurpose pins' paths (SP4T or SPDT switches) to digital mode, whose signal path is FPGA (PL) -> Level Translator -> switch -> 90 pin connector. The IO expanders are also programmed to set the level shifters into output mode. Afterward, the IO pins will turn on and off one at a time, with a delay of the runtime amount, in milliseconds, ending with all IO pins being turned on and then back off.

- 1) When prompted to input a test mode on the DRC console, type "2" and press enter.
- 2) You will be prompted to enter a delay. Each pin will toggle HIGH, then LOW. This delay is the delay, in ms, between testing each pin. A value of **500-2000** is sufficient. Enter the delay. Do not press enter yet.
- 3) On the Test Board Console, press "2". The Test Board will be configured for mode 2 without pressing enter.
- 4) Press enter on the DRC console.
- 5) You will now see this:



COM14 - Tera Term VT

```
File Edit Setup Control Window KanjiCode Help
Test Mode 2: Digital Output Test.
Set all multi-function pins to digital mode and cycles IO pins.
Delay: 100 milliseconds

Pin 2: 0      Pin 32: 0      Pin 62: 0
Pin 3: 0      Pin 33: 0      Pin 63: 0
Pin 4: 0      Pin 34: 1      Pin 64: 0
Pin 5: 0      Pin 35: 0      Pin 65: 0
Pin 6: 0      Pin 36: 0      Pin 66: 0
Pin 7: 0      Pin 37: 0      Pin 67: 0
Pin 8: 0      Pin 38: 0      Pin 68: 0
Pin 9: 0      Pin 39: 0      Pin 69: 0
Pin 10: 0     Pin 40: 0      Pin 70: 0
Pin 11: 0     Pin 41: 0      Pin 71: 0
Pin 12: 0     Pin 42: 0      Pin 72: 0
Pin 13: 0     Pin 43: 0      Pin 73: 0
Pin 14: 0     Pin 44: 0      Pin 74: 0
Pin 15: 0     Pin 45: 0      Pin 75: 0
Pin 16: 0     Pin 46: 0      Pin 76: 0
Pin 17: 0     Pin 47: 0      Pin 77: 0
Pin 18: 0     Pin 48: 0      Pin 78: 0
Pin 19: 0     Pin 49: 0      Pin 79: 0
Pin 20: 0     Pin 50: 0      Pin 80: 0
Pin 21: 0     Pin 51: 0      Pin 81: 0
Pin 22: 0     Pin 52: 0      Pin 82: 0
Pin 23: 0     Pin 53: 0      Pin 83: 0
Pin 24: 0     Pin 54: 0      Pin 84: 0
Pin 25: 0     Pin 55: 0      Pin 85: 0
Pin 26: 0     Pin 56: 0      Pin 86: 0
Pin 27: 0     Pin 57: 0      Pin 87: 0
Pin 28: 0     Pin 58: 0      Pin 88: 0
Pin 29: 0     Pin 59: 0      Pin 89: 0
```

COM17 - Tera Term VT

```
File Edit Setup Control Window KanjiCode Help
DRC Rev. 2.0 Device Verification Test.
Device: STM32H756ZIT6 Test Board.
Press escape to return to menu.

2) DRC Digital Output Test.
The following shows the current output state of the DRC, as read by the interposer.

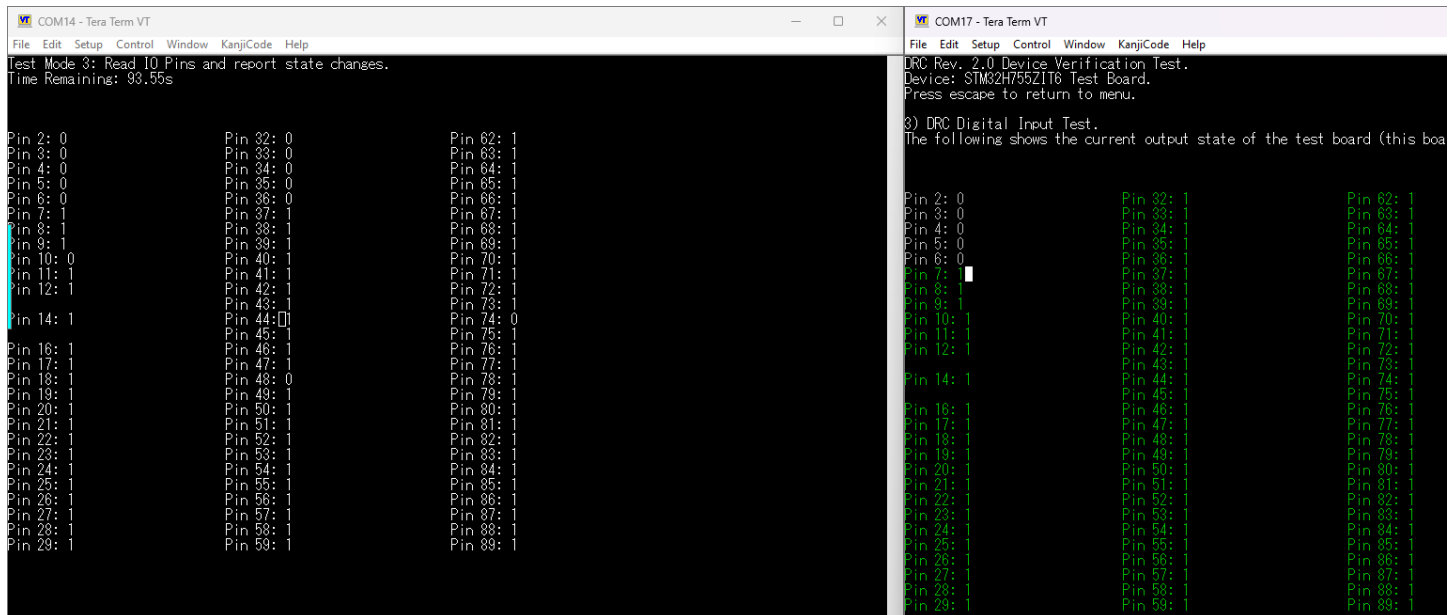
Pin 2: 0      Pin 32: 0      Pin 62: 1
Pin 3: 0      Pin 33: 0      Pin 63: 1
Pin 4: 1      Pin 34: 0      Pin 64: 1
Pin 5: 1      Pin 35: 1      Pin 65: 1
Pin 6: 0      Pin 36: 0      Pin 66: 1
Pin 7: 1      Pin 37: 1      Pin 67: 1
Pin 8: 1      Pin 38: 1      Pin 68: 1
Pin 9: 1      Pin 39: 1      Pin 69: 1
Pin 10: 0     Pin 40: 1      Pin 70: 1
Pin 11: 1     Pin 41: 1      Pin 71: 1
Pin 12: 1     Pin 42: 1      Pin 72: 1
Pin 13: 1     Pin 43: 1      Pin 73: 1
Pin 14: 1     Pin 44: 1      Pin 74: 0
Pin 15: 1     Pin 45: 1      Pin 75: 1
Pin 16: 1     Pin 46: 1      Pin 76: 1
Pin 17: 1     Pin 47: 1      Pin 77: 1
Pin 18: 1     Pin 48: 0      Pin 78: 1
Pin 19: 1     Pin 49: 1      Pin 79: 1
Pin 20: 1     Pin 50: 1      Pin 80: 1
Pin 21: 1     Pin 51: 1      Pin 81: 1
Pin 22: 1     Pin 52: 1      Pin 82: 1
Pin 23: 1     Pin 53: 1      Pin 83: 1
Pin 24: 1     Pin 54: 1      Pin 84: 1
Pin 25: 1     Pin 55: 1      Pin 85: 1
Pin 26: 1     Pin 56: 1      Pin 86: 1
Pin 27: 1     Pin 57: 1      Pin 87: 1
Pin 28: 1     Pin 58: 1      Pin 88: 1
Pin 29: 1     Pin 59: 1      Pin 89: 1
```

- 6) Verify, on the Test Board console, that each pin is transitioning from HIGH to LOW, and from LOW to HIGH. If any pin fails to change state in its turn, the digital output test fails.
- 7) Verify that the Test Board status LEDs indicate **0010**.
- 8) If all IO pins are functional, the DRC outputs have been successfully verified.
- 9) The DRC Console will automatically return to the menu. Press escape on the Test Board console to return to the menu.

Test Mode 3: Read IO Pins and report state changes.

This test will first set all multipurpose pins' paths (SP4T or SPDT switches) to digital mode, whose signal path is 90 pin connector -> switch -> Level translator -> FPGA (PL). The IO expanders are also programmed to set the level shifters into input mode. Afterward, the DRC will read each IO pin state on the 90 pin connector, and report the state of all pins.

- 1) When prompted to input a test mode on the DRC console, type "3" and press enter.
- 2) You will be prompted to enter a runtime. This is the length, in seconds, in which the DRC will probe the input pins. A value of 10-60s is sufficient. Do not press enter yet.
- 3) On the Test Board Console, press "3". The Test Board will be configured for mode 3 without pressing enter.
- 4) Press enter on the DRC console.
- 5) The following should be shown on the consoles(Will see some variation).



```

COM14 - Tera Term VT
File Edit Setup Control Window KanjiCode Help
Test Mode 3: Read IO Pins and report state changes.
Time Remaining: 93.55s

Pin 2: 0      Pin 32: 0      Pin 62: 1
Pin 3: 0      Pin 33: 0      Pin 63: 1
Pin 4: 0      Pin 34: 0      Pin 64: 1
Pin 5: 0      Pin 35: 0      Pin 65: 1
Pin 6: 0      Pin 36: 0      Pin 66: 1
Pin 7: 1      Pin 37: 1      Pin 67: 1
Pin 8: 1      Pin 38: 1      Pin 68: 1
Pin 9: 1      Pin 39: 1      Pin 69: 1
Pin 10: 0     Pin 40: 1      Pin 70: 1
Pin 11: 1     Pin 41: 1      Pin 71: 1
Pin 12: 1     Pin 42: 1      Pin 72: 1
Pin 13: 1     Pin 43: 1      Pin 73: 1
Pin 14: 1     Pin 44: 1      Pin 74: 0
Pin 15: 1     Pin 45: 1      Pin 75: 1
Pin 16: 1     Pin 46: 1      Pin 76: 1
Pin 17: 1     Pin 47: 1      Pin 77: 1
Pin 18: 1     Pin 48: 0      Pin 78: 1
Pin 19: 1     Pin 49: 1      Pin 79: 1
Pin 20: 1     Pin 50: 1      Pin 80: 1
Pin 21: 1     Pin 51: 1      Pin 81: 1
Pin 22: 1     Pin 52: 1      Pin 82: 1
Pin 23: 1     Pin 53: 1      Pin 83: 1
Pin 24: 1     Pin 54: 1      Pin 84: 1
Pin 25: 1     Pin 55: 1      Pin 85: 1
Pin 26: 1     Pin 56: 1      Pin 86: 1
Pin 27: 1     Pin 57: 1      Pin 87: 1
Pin 28: 1     Pin 58: 1      Pin 88: 1
Pin 29: 1     Pin 59: 1      Pin 89: 1

COM17 - Tera Term VT
File Edit Setup Control Window KanjiCode Help
DRC Rev. 2.0 Device Verification Test.
Device: STM32H755ZIT6 Test Board.
Press escape to return to menu.

3) DRC Digital Input Test.
The following shows the current output state of the test board (this board)

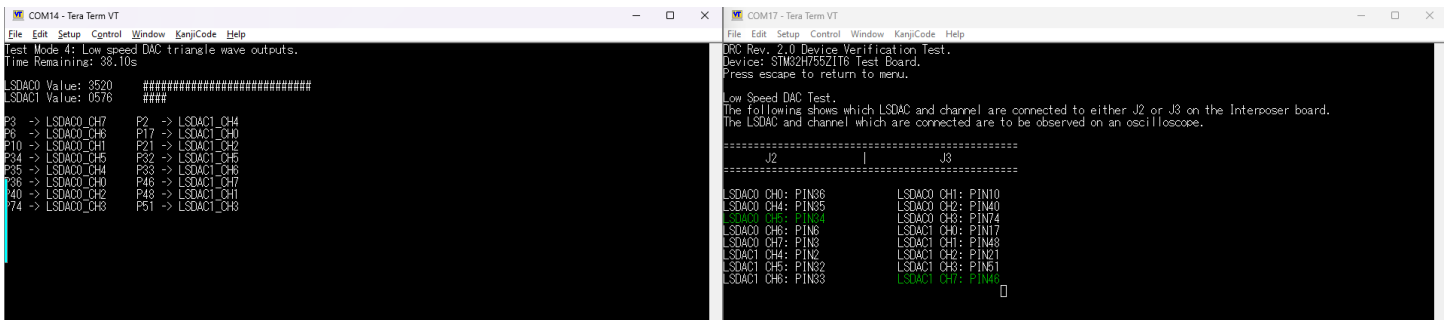
Pin 2: 0      Pin 32: 1      Pin 62: 1
Pin 3: 0      Pin 33: 1      Pin 63: 1
Pin 4: 0      Pin 34: 1      Pin 64: 1
Pin 5: 0      Pin 35: 1      Pin 65: 1
Pin 6: 0      Pin 36: 1      Pin 66: 1
Pin 7: 1      Pin 37: 1      Pin 67: 1
Pin 8: 1      Pin 38: 1      Pin 68: 1
Pin 9: 1      Pin 39: 1      Pin 69: 1
Pin 10: 1     Pin 40: 1      Pin 70: 1
Pin 11: 1     Pin 41: 1      Pin 71: 1
Pin 12: 1     Pin 42: 1      Pin 72: 1
Pin 13: 1     Pin 43: 1      Pin 73: 1
Pin 14: 1     Pin 44: 1      Pin 74: 1
Pin 15: 1     Pin 45: 1      Pin 75: 1
Pin 16: 1     Pin 46: 1      Pin 76: 1
Pin 17: 1     Pin 47: 1      Pin 77: 1
Pin 18: 1     Pin 48: 1      Pin 78: 1
Pin 19: 1     Pin 49: 1      Pin 79: 1
Pin 20: 1     Pin 50: 1      Pin 80: 1
Pin 21: 1     Pin 51: 1      Pin 81: 1
Pin 22: 1     Pin 52: 1      Pin 82: 1
Pin 23: 1     Pin 53: 1      Pin 83: 1
Pin 24: 1     Pin 54: 1      Pin 84: 1
Pin 25: 1     Pin 55: 1      Pin 85: 1
Pin 26: 1     Pin 56: 1      Pin 86: 1
Pin 27: 1     Pin 57: 1      Pin 87: 1
Pin 28: 1     Pin 58: 1      Pin 88: 1
Pin 29: 1     Pin 59: 1      Pin 89: 1
  
```

- 6) Verify that the DRC status LEDs indicate 0011.
- 7) Verify that the Test Board status LEDs indicate 0011.
- 8) The Test Board will now output different states on the IO pins and displays its outputs on the Test Board console. The DRC will read these pins and display the states read on the DRC console. Verify that both consoles show the same activity and are synchronized.
- 9) If the test succeeds, the DRC inputs have been successfully verified.
- 10) Press escape on the Test Board console to go back to the menu. The DRC console will time out after the length of the test has passed.

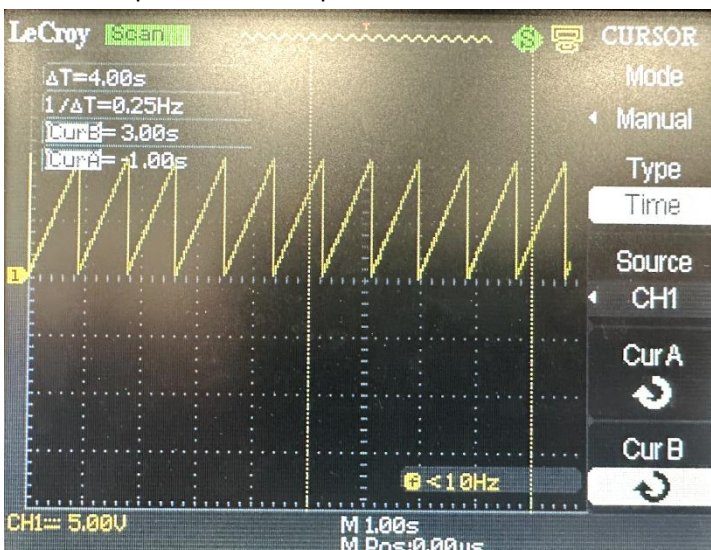
Test Mode 4: Low speed DAC sawtooth wave outputs.

This test will first set all multipurpose pins' paths (SP4T or SPDT switches) to Low Speed DAC mode, whose signal path is Low Speed DAC channel -> switch -> 90 pin connector. Afterward, the DRC will send commands to all 16 Low Speed DAC channels to output a low frequency sawtooth wave.

- 1) Type "4" on the Test Board console to enter into mode 4.
- 2) Type "4" on the DRC console and press enter.
- 3) You will be prompted to enter a runtime. This is the length, in seconds, in which the DRC will output the LSDAC values. Test cannot be changed until this test times out. A value of 10-60 is recommended.
- 4) The DRC console will show the LSDAC connections and the current value of each LSDAC set. The Test Board will route each LSDAC output to either J2 or J3, so that the signals are viewable on an oscilloscope. The Test Board console will show which LSDAC channel is currently connected, in green.



- 5) Verify that the DRC status LEDs indicate 0100.
- 6) that the Test Board status LEDs indicate 0100.
- 7) Using an oscilloscope, verify that a low frequency sawtooth wave is generated on each pin as its turn in the sequence comes up.

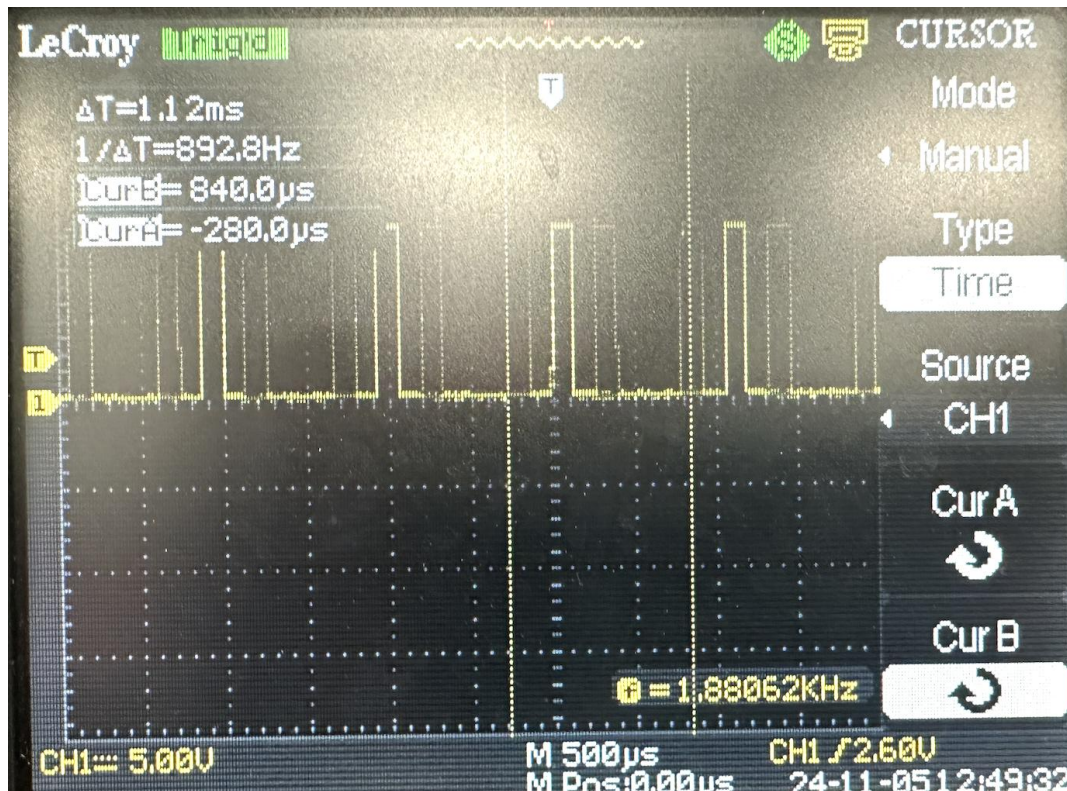


- 8) This test is meant to measure linearity, as well as min/max voltages. Verify 0-10V and a linear slope. Firmware may have different sawtooth wave, but checking for linearity and voltage range is the requirement.
- 9) If all Low Speed DAC channels have been verified, move on to test mode 5.

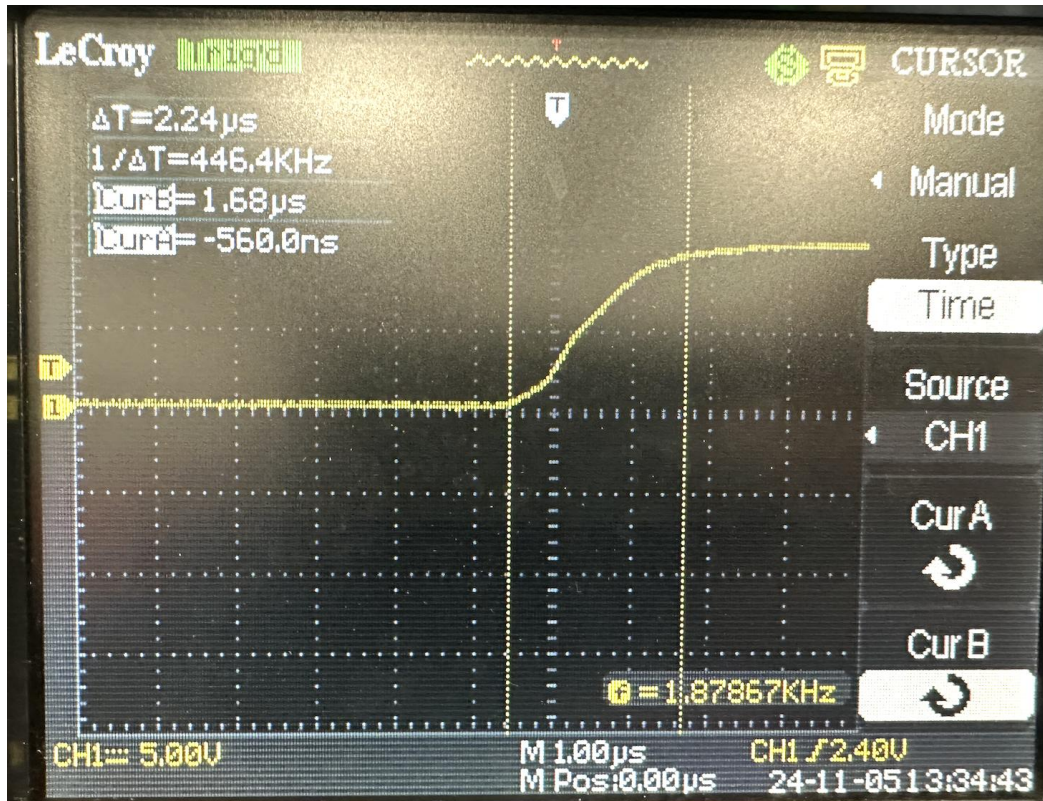
Test Mode 5: Low speed DAC square wave outputs. Transition time test.

This test will first set all multipurpose pins' paths (SP4T or SPDT switches) to Low Speed DAC mode, whose signal path is Low Speed DAC channel -> switch -> 90 pin connector. Afterward, the DRC will send commands to all 16 Low Speed DAC channels to cycle between 0V and 10V.

- 1) Type "5" on the Test Board console to enter into mode 5.
- 2) Type "5" on the DRC console and press enter.
- 3) You will be prompted to enter a runtime. This is the length, in seconds, in which the DRC will output the LSDAC values. Test cannot be changed until this test times out. A value of 10-60 is recommended.
- 4) The DRC console will show the LSDAC connections and the current value of each LSDAC set. The Test Board will route each LSDAC output to either J2 or J3, so that the signals are viewable on an oscilloscope. The Test Board console will show which LSDAC channel is currently connected, in green.
- 5) Verify that the DRC status LEDs indicate **0101**.
- 6) Using an oscilloscope, verify that a square wave is generated on each pin that is set to Low Speed DAC mode (16 pins total). This the switching between LSDAC channels will be identical to the method shown in test mode 4.



- 1) On each channel, zoom in and verify the transition rate is sharp as seen below.

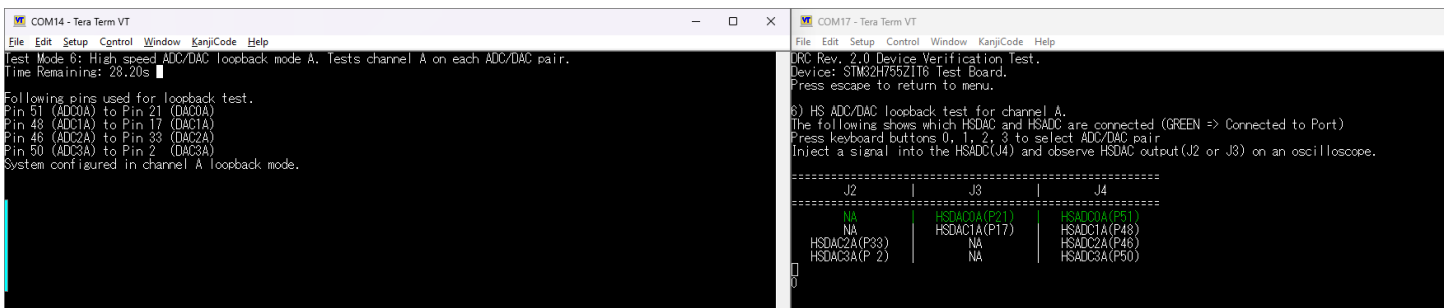


- 11) If all Low Speed DAC channels have been verified, move on to test mode 6.

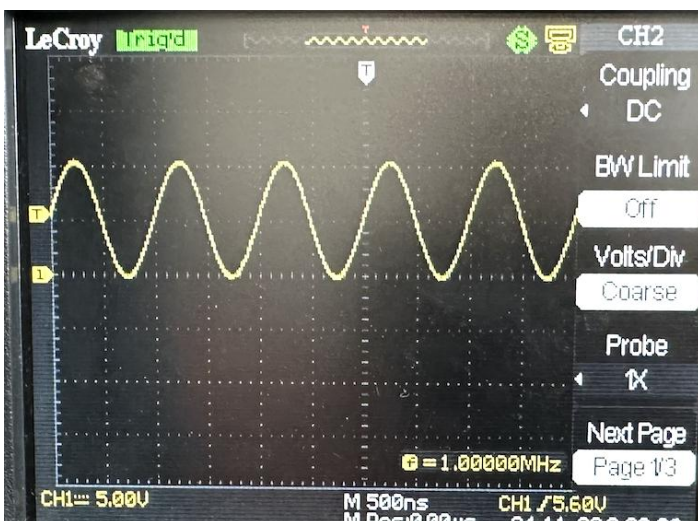
Test Mode 6: High speed ADC/DAC loopback mode A.

Tests channel A on each ADC/DAC pair. Since some High Speed ADC and High Speed DAC channels share a multifunction pin, testing channel A and B for the converters will be done in separate tests.

- 1) Connect a function generator into J4 on the DVT interposer and set the frequency to **1MHz sine wave, at 10Vpp, with a 5V offset**. In other words, a 1MHz signal from 0-10V.
- 2) Connect J2 and J3 to an oscilloscope as done in previous tests.
- 3) Type **"6"** on the Test Board console to enter into mode 6.
- 4) Type **"6"** on the DRC console and press enter.
- 5) You will be prompted to enter a runtime. This is the length, in seconds, in which the DRC will remain in loopback mode. Test cannot be changed until this test times out. A value of 10-60 is recommended. Press enter.
- 6) Verify that the DRC status LEDs indicate **0110**.
- 7) that the Test Board status LEDs indicate **0110**.
- 8) The following output should be observed:



- 9) On the Test Board console, pressing **0, 1, 2, 3** will highlight a different row, in green. The selected row will indicate which connections are made. For example, pressing **0**, as seen above, connects **HSADC0A(P51)** to **J4**, or the function generator, and **HSDAC0A(P21)** to **J3**, or the oscilloscope channel.
- 10) Go through each row, and verify that the scope shows the DRC outputting a **1MHz sine wave from -5V to 5V**.

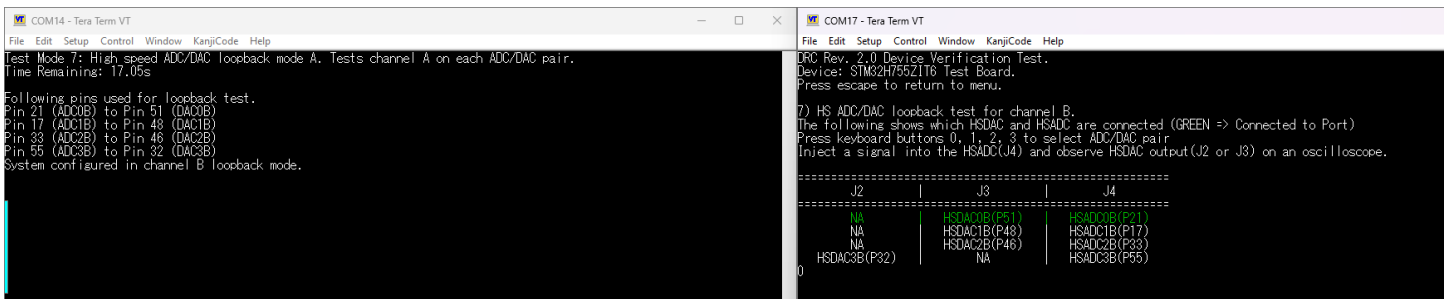


- 11) Once all Channel A ADC/DAC pairs are verified, move on to test mode 7.

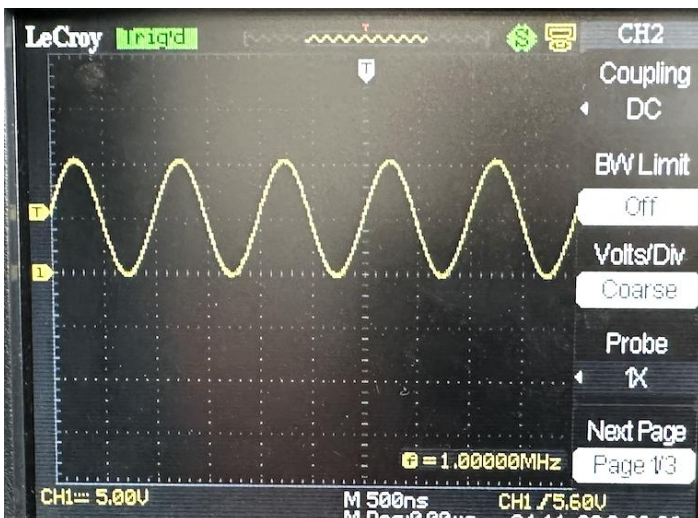
Test Mode 7: High speed ADC/DAC loopback mode B.

Tests channel B on each ADC/DAC pair. Since some High Speed ADC and High Speed DAC channels share a multifunction pin, testing channel A and B for the converters will be done in separate tests.

- 1) Connect a function generator into J4 on the DVT interposer and set the frequency to **1MHz sine wave, at 10Vpp, with a 5V offset**. In other words, a 1MHz signal from 0-10V.
- 2) Connect J2 and J3 to an oscilloscope as done in previous tests.
- 3) Type **"7"** on the Test Board console to enter into mode 7.
- 4) Type **"7"** on the DRC console and press enter.
- 5) You will be prompted to enter a runtime. This is the length, in seconds, in which the DRC will remain in loopback mode. Test cannot be changed until this test times out. A value of 10-60 is recommended. Press enter.
- 6) Verify that the DRC status LEDs indicate **0111**.
- 7) that the Test Board status LEDs indicate **0111**.
- 8) The following output should be observed:



- 9) On the Test Board console, pressing **0, 1, 2, 3** will highlight a different row, in green. The selected row will indicate which connections are made. For example, pressing **0**, as seen above, connects **HSADC0B(P21)** to **J4**, or the function generator, and **HSDAC0B(P51)** to **J3**, or the oscilloscope channel.
- 10) Go through each row, and verify that the scope shows the DRC outputting a **1MHz sine wave from -5V to 5V**.

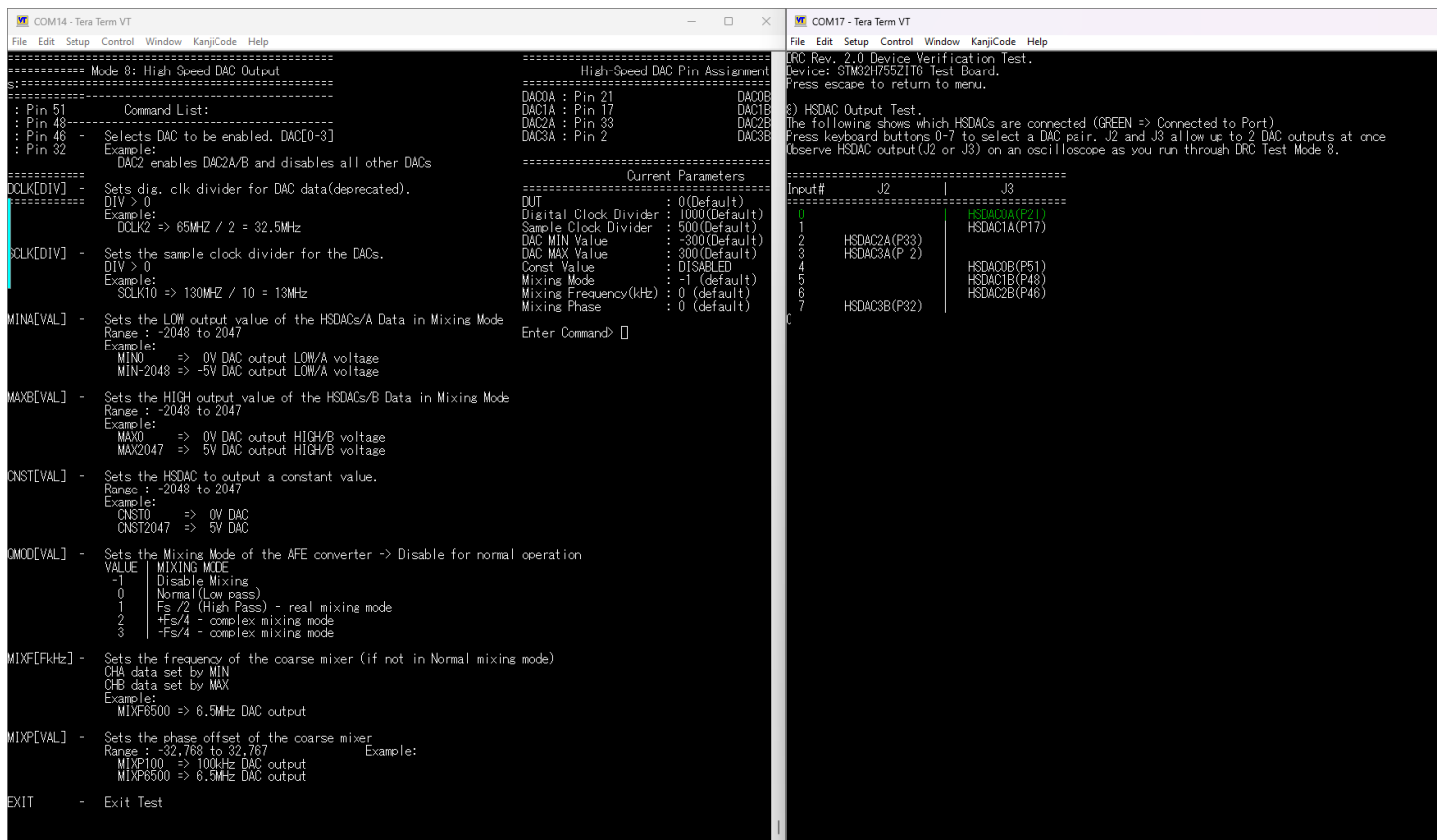


- 11) Once all Channel B ADC/DAC pairs are verified, move on to test mode 8.

Test Mode 8: High speed DAC Output and Quadrature Modulation Test

This mode will test the HSDAC channels. There is the flexibility to change amplitude, frequency, and modulation modes.

- 1) Connect J2 and J3 to an oscilloscope as done in previous tests.
- 2) Type “8” on the Test Board console to enter into mode 8.
- 3) Type “8” on the DRC console and press enter.
- 4) Verify that the DRC status LEDs indicate 1000.
- 5) that the Test Board status LEDs indicate 1000.
- 6) The following output should be observed:



```

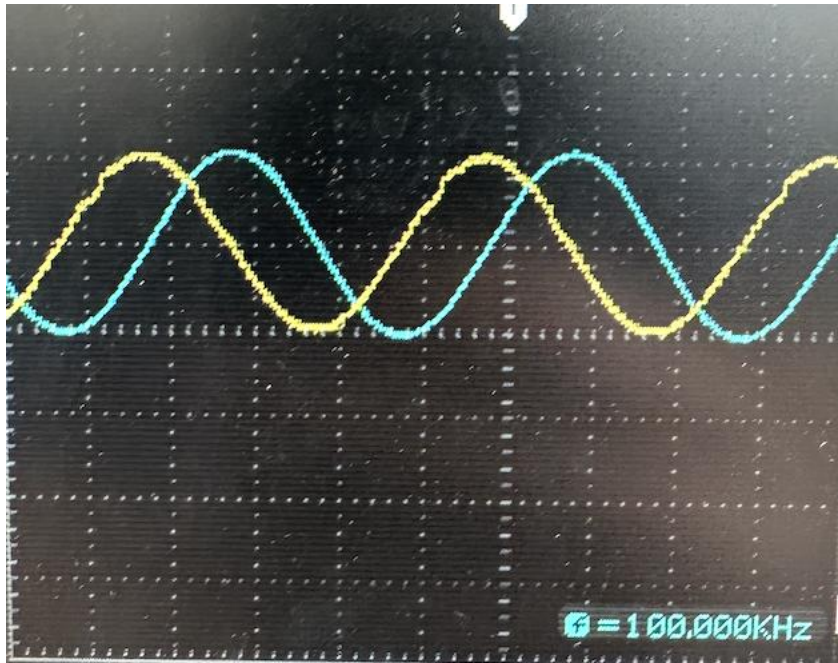
COM14 - Tera Term VT
File Edit Setup Control Window KanjiCode Help
===== Mode 8: High Speed DAC Output =====
: Pin 51 Command List:
: Pin 48 - Selects DAC to be enabled. DAC[0-3]
: Pin 46 Example:
: Pin 32 DAC2 enables DAC2A/B and disables all other DACs
=====
DCLK[DIV] - Sets djs. clk divider for DAC data(deprecated).
Range: DIV > 0
Example:
DCLK2 => 65MHz / 2 = 32.5MHz
=====
SCLK[DIV] - Sets the sample clock divider for the DACs.
Range: DIV > 0
Example:
SCLK10 => 130MHz / 10 = 13MHz
=====
MINA[VAL] - Sets the LOW output value of the HSDACs/A Data in Mixing Mode
Range: -2048 to 2047
Example:
MIN0 => 0V DAC output LOW/A voltage
MIN-2048 => -5V DAC output LOW/A voltage
=====
MAXB[VAL] - Sets the HIGH output value of the HSDACs/B Data in Mixing Mode
Range: -2048 to 2047
Example:
MAX0 => 0V DAC output HIGH/B voltage
MAX2047 => 5V DAC output HIGH/B voltage
=====
CNST[VAL] - Sets the HSDAC to output a constant value.
Range: -2048 to 2047
Example:
CNST0 => 0V DAC
CNST2047 => 5V DAC
=====
QMOD[VAL] - Sets the Mixing Mode of the AFE converter -> Disable for normal operation
Range: -1 to 3
Example:
-1 Disable Mixing
0 Normal(Low pass)
1 Fs/2 (High Pass) - real mixing mode
2 +Fs/4 - complex mixing mode
3 -Fs/4 - complex mixing mode
=====
MIXF[FkHz] - Sets the frequency of the coarse mixer (if not in Normal mixing mode)
Range: CHA data set by MIN
CHB data set by MAX
Example:
MIXF6500 => 6.5MHz DAC output
=====
MIXP[VAL] - Sets the phase offset of the coarse mixer
Range: -32,768 to 32,767
Example:
MIXP100 => 100kHz DAC output
MIXP6500 => 6.5MHz DAC output
=====
EXIT - Exit Test

COM17 - Tera Term VT
File Edit Setup Control Window KanjiCode Help
DRC Rev. 2.0 Device Verification Test.
Device: STM32H755ZIT6 Test Board.
Press escape to return to menu.
8) HSDAC Output Test.
The following shows which HSDACs are connected (GREEN => Connected to Port)
Press keyboard buttons 0-7 to select a DAC pair, J2 and J3 allow up to 2 DAC outputs at once
Observe HSDAC output(J2 or J3) on an oscilloscope as you run through DRC Test Mode 8.
=====
Input# J2 J3
=====
0 HSDAC0A(P21) HSDAC0A(P17)
1 HSDAC0B(P33) HSDAC0B(P17)
2 HSDAC2A(P33) HSDAC2A(P17)
3 HSDAC3A(P 2) HSDAC3A(P17)
4 HSDAC0B(P51) HSDAC0B(P48)
5 HSDAC1B(P48) HSDAC1B(P48)
6 HSDAC2B(P48) HSDAC2B(P48)
7 HSDAC3B(P32) HSDAC3B(P32)
0

```

- 7) On the Test Board console, pressing 0 through 7 will highlight a different row, in green. The selected row will indicate which connections are made. For example, pressing 0, as seen above, connects HSDAC0A(P21) to J3, or the, or the oscilloscope channel.
- 8) On the DRC console, there are different commands that can be used, and their usage is listed in the console output. The DRC will only enable a single converters frontend at a time, and the command “DAC[AFENUM] must be used to switch between Converters. DAC0 corresponds to HSDAC0A and HSDAC0B.
- 9) To verify output capability, use the following commands in the DRC console in order:
 - a. QMOD0
 - b. SCLK2
 - c. CNST2047
 - d. MIXF1000
 - e. DAC0

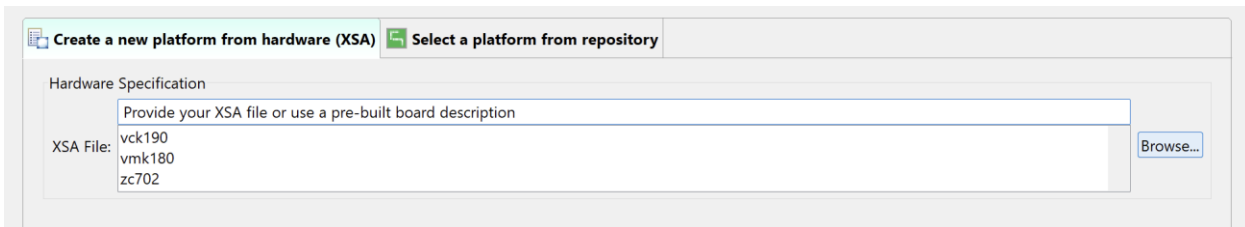
- f. DAC1
 - g. DAC2
 - h. DAC3
- 10) For each of the above DAC# commands, make sure to route the DAC output using the Test Board console. For example, after running DAC0, use rows 0 and 4 to view the DAC outputs on the scope.
 - 11) Once the first 4 commands are sent, you can switch between DACs without having to redo the setup.
 - 12) Run the command "DAC2" on the DRC console, and input "2" and "6" on the Test Board console.
 - 13) Converter 2 is the only Converter in which Channel A and B are on different scope channels. You should see a quadrature modulated output, where there are 2 sine waves 90 degrees apart.



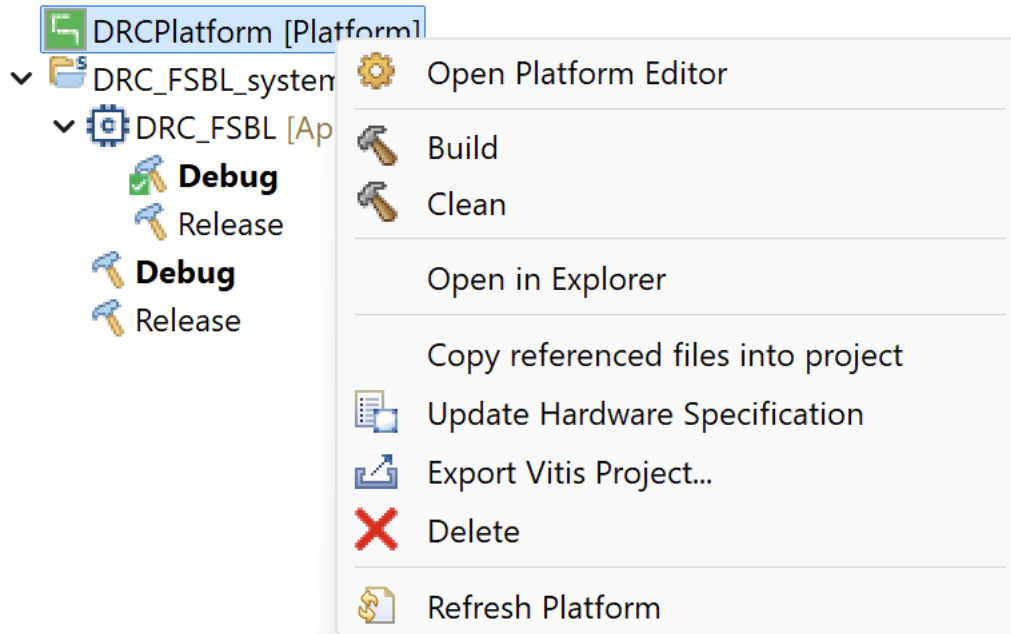
In the above, MIXF100 was used instead of MIXF1000, thus 100kHz.

Creating a Project

- 1) Prepare the XSA file using Vivado 2023.1.
 - a. NOTE: When creating the Vivado Project, use the Octavo supplied BDF for the SiP w/ 1.8V QSPI flash.
 - b. NOTE: When placing the Zynq processor block in Vivado, run board automation and make sure the apply presets option is checked.
 - c. NOTE: Double click on the Zynq block and disable all gigabit transceivers. The preset enables GEM, SATA and will cause the FSBL to fail.
- 2) Open Vitis 2023.1 and create a platform using said XSA file.
 - a. File->New->Platform Project...
- 3) Name the platform and click on “Next >”
- 4) You will see this page. Browse for your XSA file.

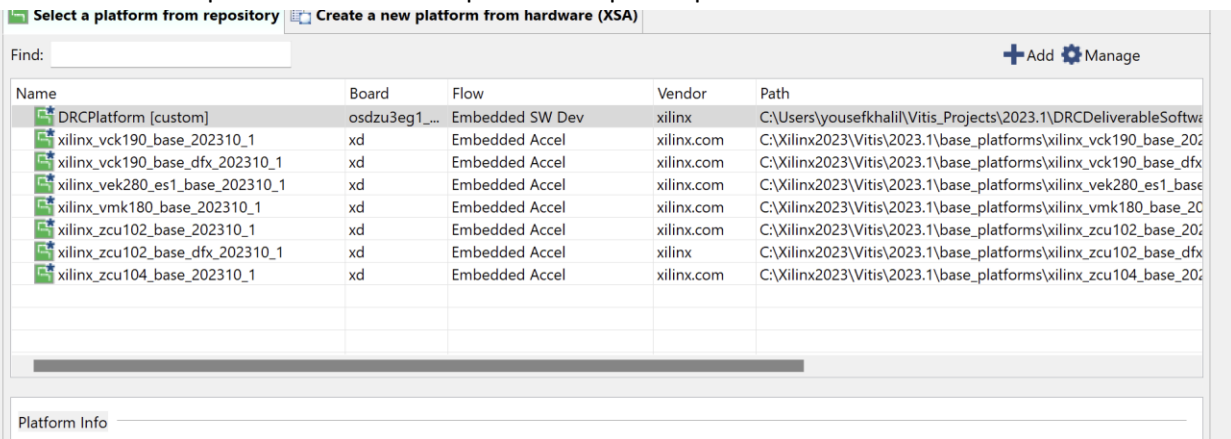


- 5) Keep all settings unchanged and click on “Finish”
 - a. Operating System is “standalone”
 - b. Processor should show the A53.
- 6) Build the platform (Note: You will only see the DRCPlatform [Platform] at this stage)



- 7) Once built, you will create an application project
 - a. File ->New-> Application Project...

8) Select the custom platform made in the previous steps and press “Next >”



9) Name the application and proceed. The domain on the next page will be the default.

10) Select either an empty application or Hello World template.

11) Press “Finish”

12) You will now create another Application Project

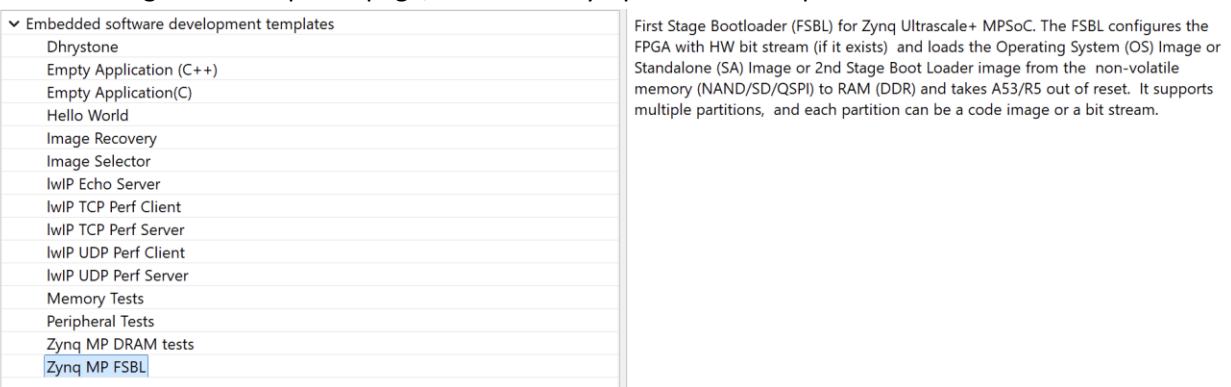
a. File->New->Application Project...

13) Select the custom platform just as in the previous project.

14) This project will become the FSBL loaded for booting off QSPI flash. Name the application in a way that it is clear to be the FSBL.

15) Continue. When on the domain page, select new domain, and name it to “fsbl_domain” instead of “standalone_domain.”

16) When arriving at the templates page, select the “Zynq MP FSBL” template.



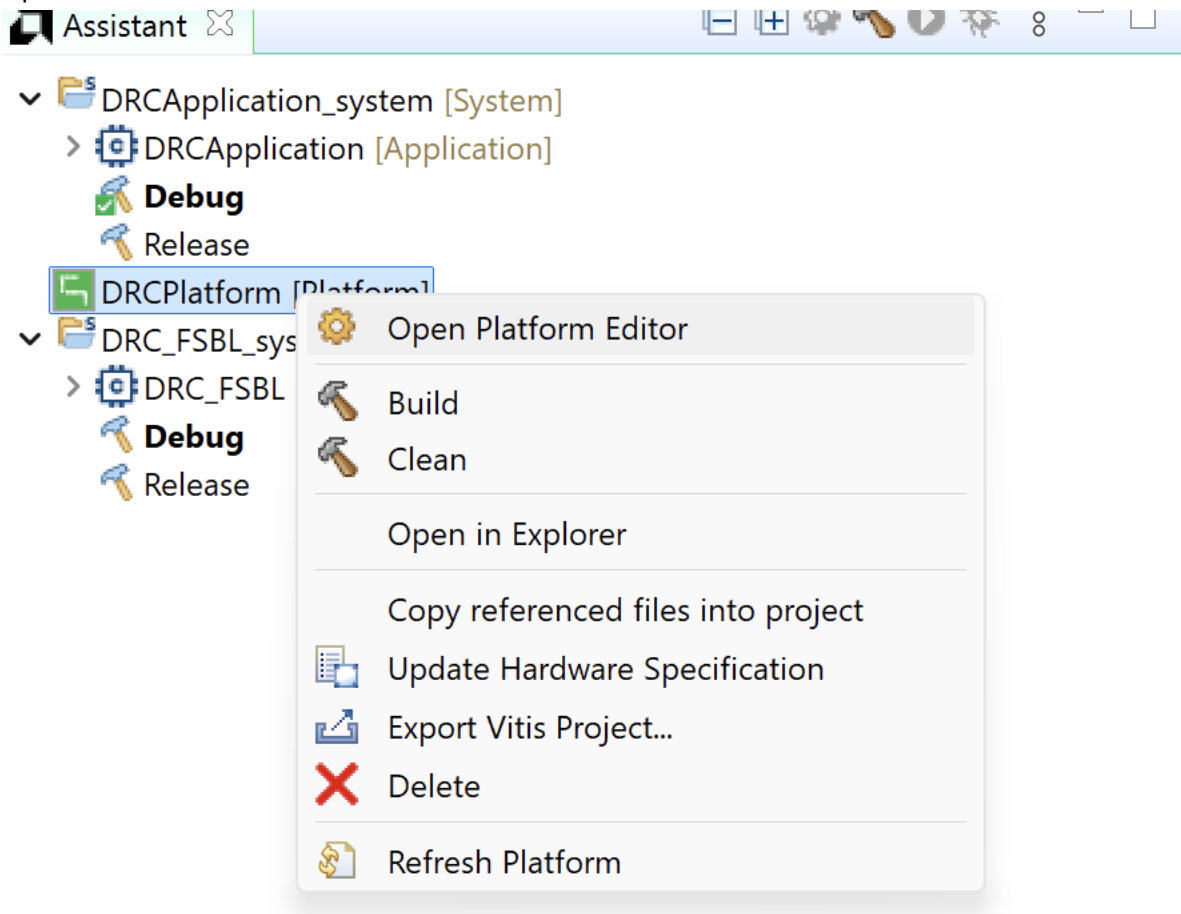
17) Press “Finish” and wait for Vitis to complete its task.

18) Once complete, your file structure should be similar to this:

- ✓  DRCApplication_system [System]
 - >  DRCApplication [Application]
 -  **Debug**
 -  Release
 -  DRCPlatform [Platform]
- ✓  DRC_FSBL_system [System]
 - >  DRC_FSBL [Application]
 -  **Debug**
 -  Release

There should be 1 platform, and 2 system projects, with 1 application project in each.

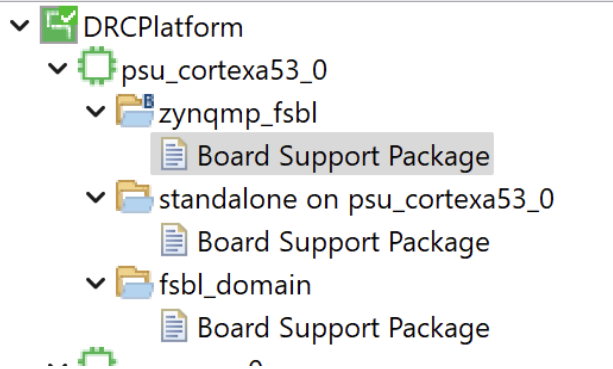
19) Open the Platform Editor.



The screenshot shows the Assistant window with the Assistant tab selected. The project tree on the left is the same as in the previous image. The DRCPlatform [Platform] project is selected, and a context menu is open over it. The menu options are:

-  Open Platform Editor
-  Build
-  Clean
- Open in Explorer
- Copy referenced files into project
-  Update Hardware Specification
-  Export Vitis Project...
-  Delete
-  Refresh Platform

20) You should see the following:



View current BSP settings, or configure settings; assign drivers to peripherals, change versions c

Modify BSP Settings...

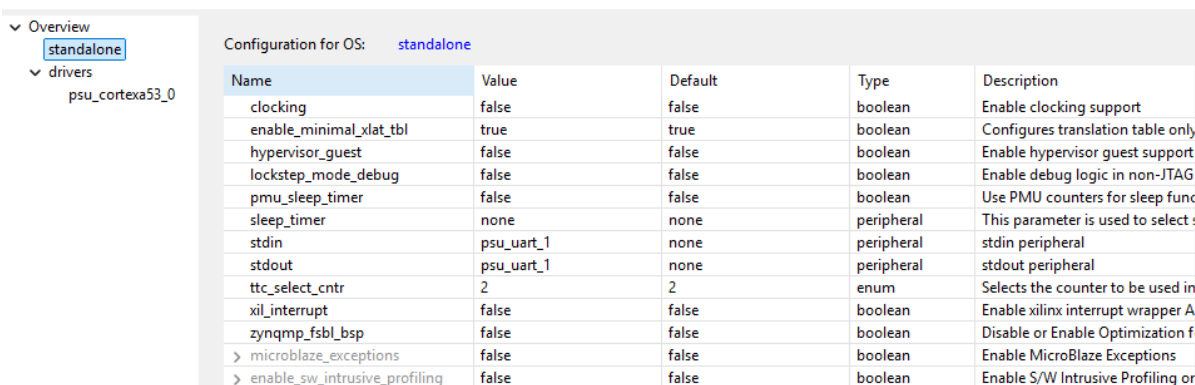
Reset BSP Sources

A BSP settings file is generated with the user o clears any existing modifications done. All the s

[Load BSP settings from file](#)

For each “Board Support Package” under the custom platform (DRCPlatform in this case), click on “Modify BSP Settings...”

21) Under standalone, change stdin/stdout to psu_coresight to allow terminal interaction through jtag.



22) Press Ok, and do this for all domains.

a. Note: Set Stdin and Stdout to psu_uart_1 to interface on the edge connector.

23) Make any changes desired in the application whose template is the hello world/empty project.

24) Build the platform.

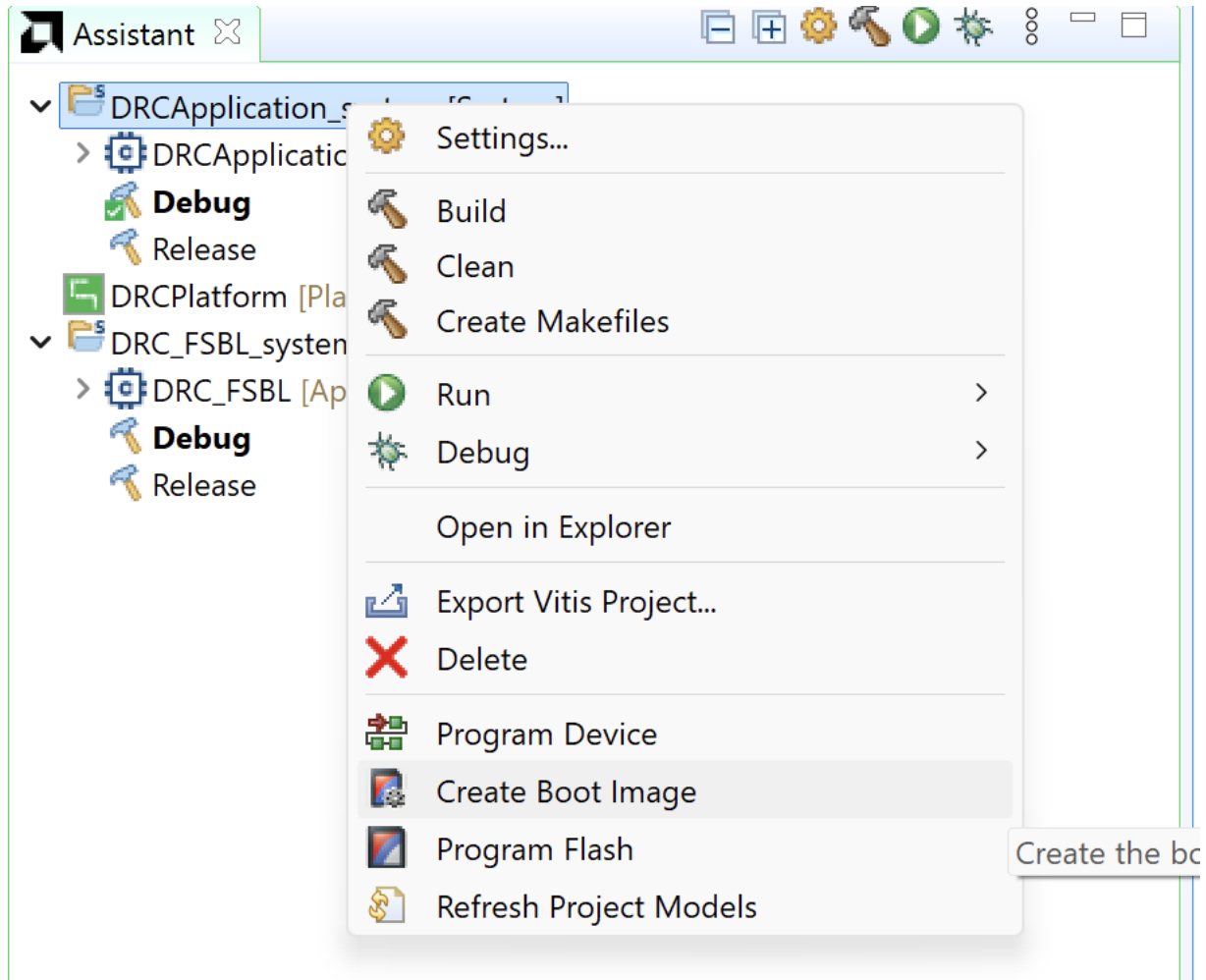
25) Build the FSBL project application.

26) Build the user application project.

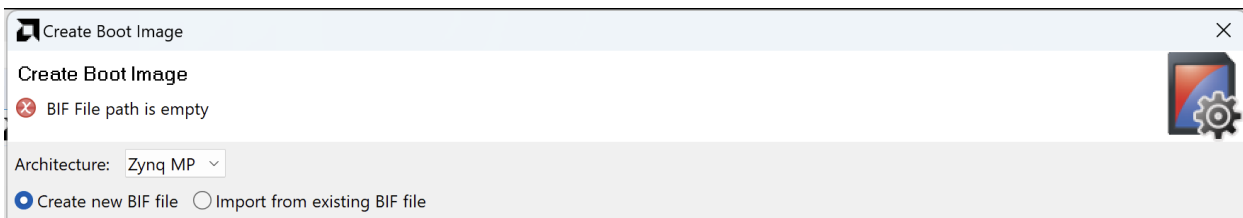
27) Continue to the next section for preparing a boot image and flashing the QSPI.

Preparing a Boot Image

- 1) Right click on the user application system project and select Create Boot Image



- 2) When the dialog appears, select “Create new BIF File,” and select Zynq MP for architecture.



3) Browse for a location to store the boot image and bif file.

Output BIF file path:		Browse...
UDF data:		Browse...
☐ Split	Output format: BIN ▼	
Output path:	C:/Xilinx2023/Vitis/2023.1/bin/BOOT.bin	Browse...

4) In the partitions:

Boot image partitions

File path	Encrypted	Authenticated

Add
Delete
Edit
Up
Down

?
Preview BIF Changes
Create Image
Cancel

Add the following:

- a. FSBL application project application.elf as the bootloader
 - i. NOTE: Do not use the fsbl generated from the user application project. The “Zynq MP FSBL” template is designed to become the FSBL when booting from a storage medium.
- b. The bitsream
- c. User application project application.elf

You should end with this:

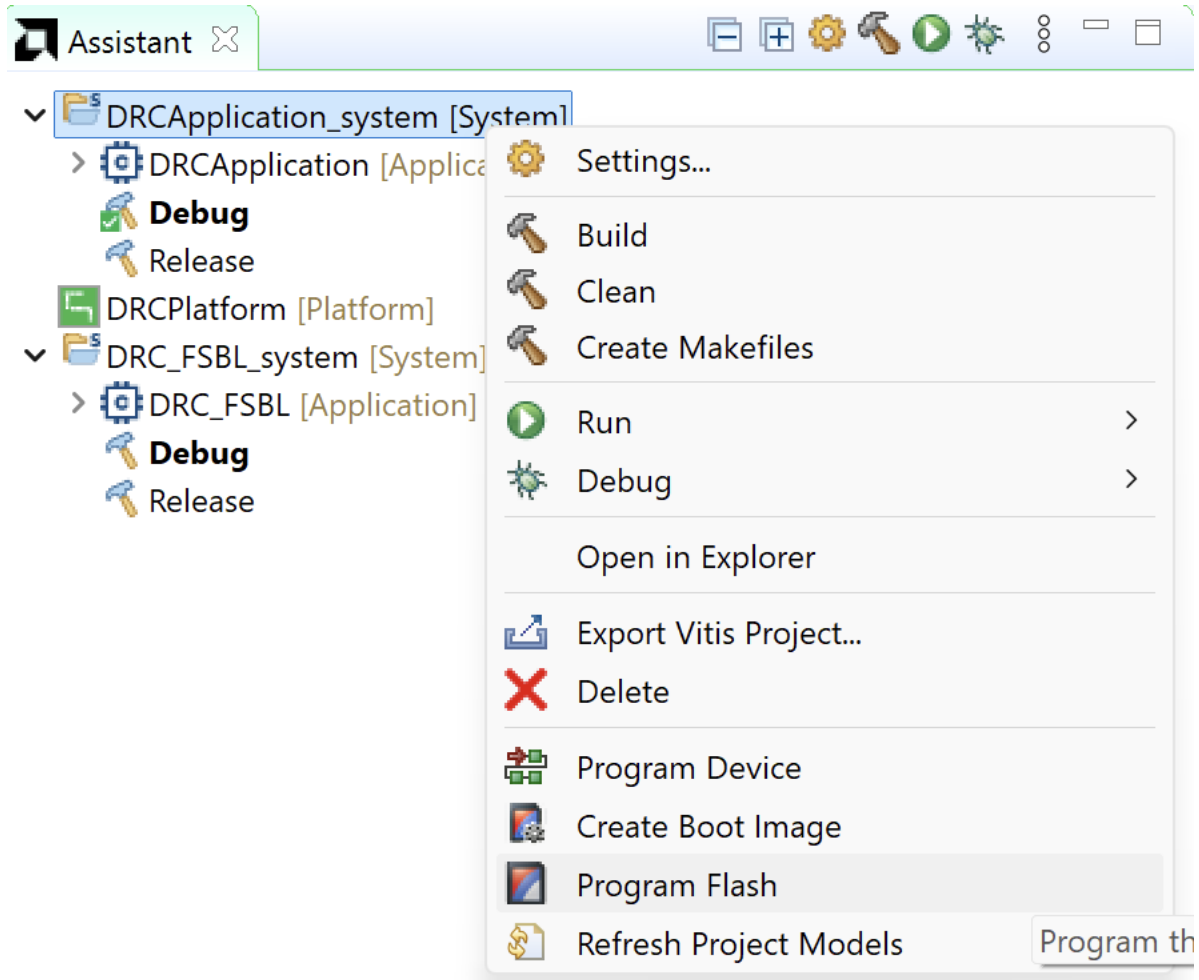
Boot image partitions

File path	Encrypted	Authenticated
(bootloader) C:\Users\yousefkhali\Vitis_Projects\2023.1\DRCDeliverableSoftware\DRC_FSBL\Debug\DRC_FSBL.elf	none	none
C:\Users\yousefkhali\Vitis_Projects\2023.1\DRCDeliverableSoftware\DRC_FSBL\ide\bitstream\design_1_wrapper.bit	none	none
C:\Users\yousefkhali\Vitis_Projects\2023.1\DRCDeliverableSoftware\DRCAApplication\Debug\DRCAApplication.elf	none	none

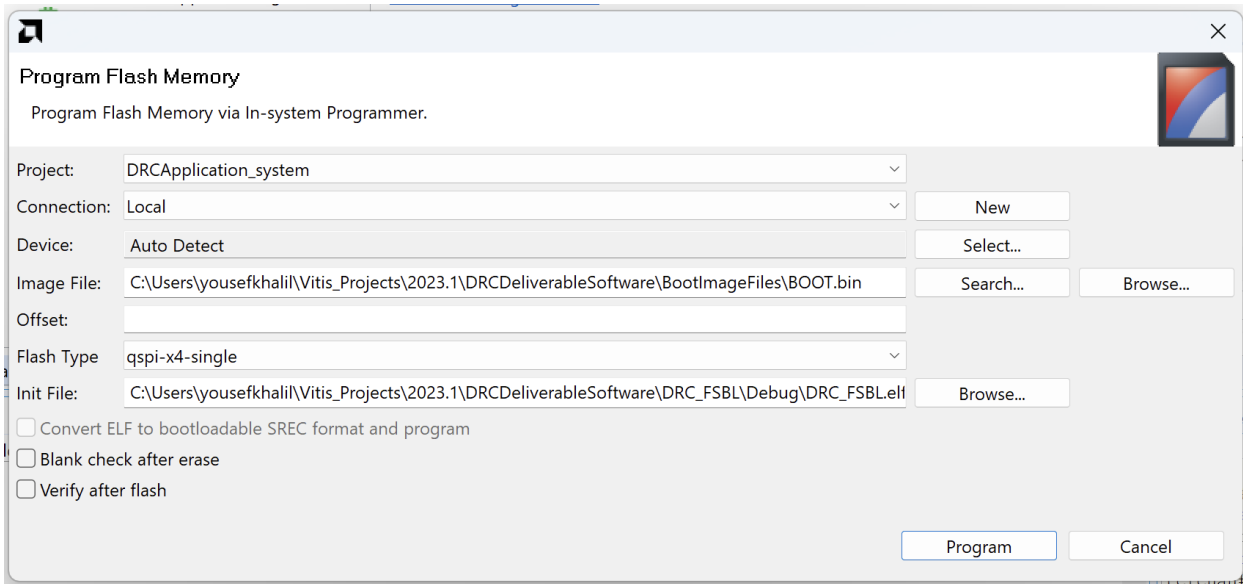
5) Click on “Create Image” to create the boot image in the output path specified.

Programming QSPI Flash

- 1) Make sure the boot mode jumper on edge connector interface board is set to JTAG boot mode :
"OFF ON OFF OFF".
- 2) Apply power to the DRC.
- 3) Connect the DRC to the PC and make sure it is showing up in the targets list in Vitis.
- 4) In the Assistant, right click on the user application system project and select "Program Flash."



5) The following page appears. Use these settings:



For Image File, browse to the location in which the BOOT.bin was generated in from the previous step. Make sure Flash type is qspi-x4-single. The init File is the FSBL application that was used before as the bootloader.

- 6) Press “Program”
- 7) Verify the device successfully programs by switching the boot jumper to QSPI “OFF OFF OFF OFF” and power cycling the system.

Additionally, the BOOT.bin file can be stored in an SD Card, and the DRC can boot of said SD card when the edge connector interface board jumpers are set to “”